Validated Dirichlet approximation and weighted arithmetic moments

GRH research working note

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Abstract

Exact outward dyadic arithmetic and theta-integral quadrature certify three finite Dirichlet readouts. Separately, a fixed-twist sextic Hecke-family moment improves an original masked remainder estimate, while retaining a positive cutoff loss. We give the precise external analytic inputs and distinguish these results from zero-location and spectral determinant claims. No proof of GRH or complete Lean formalization is asserted.

1 A finite, validated approximation result

Let $\chi_3 = (0, 1, -1)$, $\chi_4 = (0, 1, 0, -1)$ and $\chi_5 = (0, 1, -1, -1, 1)$, extended periodically, with parities $a = 1, 1, 0$. We approximate the completed functions

$$\Lambda_q(s) = (q/\pi)^{(s+a)/2}\Gamma((s+a)/2)L(s, \chi_q).$$

The primitive theta/Mellin identity is the explicit classical analytic input; no hypothesis about zeros is used. These three examples do not assert a uniform result for all conductors, general Hecke functions or Artin holomorphy.

Theorem 1 (Scope of the supplied finite certificate)**.** *For each $q = 3, 4, 5$, the supplied exact-rational calculation constructs a complex polynomial P_q of degree 28 satisfying*

$$\sup_{|s-1/2|\le 7/10} |P_q(s) - \Lambda_q(s)| < 5 \cdot 10^{-15}.$$

The computation includes enclosures of all 64 true Fourier-circle nodes, completed-function sample error, analytic truncation, Fourier aliasing, the exact closed-form 180-step gradient-descent iterate and coefficient midpoint rounding. The Python implementation and informal analytic proof have independent checks; this is not a kernel-checked Lean theorem.

Proof. Write $F(c+z) = \sum_{n\ge 0} a_n z^n$, with $c = 1/2$, $R = 7/10$, $\rho = 6/5$, $r = 12/5$. The independently recomputed rational theta majorants are strictly below $M_q = 4, 10, 6$, respectively, on $|z| \le r$. Use true roots $\zeta_j = e^{2\pi i j/64}$ and rational labels y_j with proved error $|y_j - F(c + \rho\zeta_j)| \le \eta$. Use the real Euclidean gradient on complex coefficient coordinates for

$$\mathcal{L}(b) = \frac{1}{128}\sum_{j=0}^{63}\left|\sum_{n=0}^{28} b_n\zeta_j^n - y_j\right|^2.$$

Discrete Fourier orthogonality gives the least-squares optimum $b_n^* = 64^{-1}\sum_j y_j\bar{\zeta}_j^n$, $0 \le n \le 28$, and gradient $\nabla\mathcal{L}(b) = b - b^*$. For step $1/4$ and zero initialization, the exact 180-step iterate is

$b_n^{180} = (1-(3/4)^{180})b_n^*$. The code evaluates this closed form; it does not claim an explicit180-update optimizer run.

Set $u = R/\rho = 7/12$, $v = R/r = 7/24$ and $w = (\rho/r)^{64} = 2^{-64}$. Let $\widehat{b}_n$ be an outward rectangle enclosing b_n^* and ϵ_n the sum of the coordinate halfwidths of the rectangle for b_n^{180}. Choose each actual polynomial coefficient as that rectangle's exact rational midpoint c_n. The normalized basis is explicit:

$$P_q(s) = \sum_{n=0}^{28} c_n \left(\frac{s-1/2}{6/5}\right)^n.$$

Thus the exported coefficient with index n multiplies the normalized variable, rather than the raw power $(s-1/2)^n$. Its error on the compact disk is bounded by

$$E_{\text{opt}} = (3/4)^{180} \sum_{n=0}^{28} |\widehat{b}_n|_1 u^n, \qquad E_{\text{arithmetic}} = \sum_{n=0}^{28} \epsilon_n u^n,$$

$$E_{\text{oracle}} = \eta \frac{1-u^{29}}{1-u}, \qquad E_{\text{alias}} = M_q \frac{w}{1-w} \frac{1-v^{29}}{1-v},$$

$$E_{\text{Taylor}} = M_q \frac{v^{29}}{1-v}.$$

Here $|\widehat{b}_n|_1$ is the sum of the maximum absolute real and imaginary coordinates, a safe upper bound for every enclosed complex modulus. Cauchy's estimate $|a_n| \le M_q r^{-n}$ and the Fourier identity $b_n^* = a_n\rho^n + \sum_{\ell\ge1} a_{n+64\ell}\rho^{n+64\ell}$ for exact labels prove the alias and Taylor bounds. Label perturbations contribute at most η per Fourier coefficient. The remaining two terms follow directly from the exact iterate and coefficient rectangles. Each term is rounded upward to a 180-bit dyadic rational; the reported total is the exact sum of those reported terms. The certified totals below are strictly below the asserted threshold. Detailed sample enclosures are proved in the extended edition and encoded with exact integer endpoints. □

Conductor	Maximum sample radius	Uniform polynomial bound
3	3.026×10^{-17}	1.786×10^{-15}
4	7.206×10^{-17}	4.455×10^{-15}
5	1.404×10^{-16}	2.906×10^{-15}

The displayed values are rounded upward; exact rational records establish the strict inequalities.

2 What this does and does not certify

The final JSON records exact rational centers, interval endpoints, source hashes and all five error terms. Float values are display summaries only. A high-precision comparison is an independent diagnostic, not the reason an interval contains its mathematical target. Arithmetic uses outward integer operations, and invalid float/order inputs are rejected, including through typed caches. Earlier pilots and failed validation probes are retained.

This is a fixed finite tolerance/configuration. Increasing accuracy needs new truncation and precision choices; the fixed $N = 11, T = 128$ interface is not an arbitrary-accuracy oracle. The broader approximation existence theorem is separate. The certificate says nothing about the location of zeros: P_q is not required to be real-rooted or a self-adjoint spectral determinant. The independent arithmetic determinant identity, positivity and original signed short-row/covariance estimates remain open. No GRH proof or complete new Lean4 formalization is claimed.

3 Validated primitive operations

Real intervals are closed dyadic intervals $[l2^{-224}, h2^{-224}]$ with integer endpoints; complex intervals are Cartesian products. Addition is exact at this scale. For multiplication take the minimum/maximum of all four integer endpoint products, divide by 2^{224} and round down/up. For a reciprocal whose interval omits zero, reverse the endpoint order and round $2^{448}/h$ down and $2^{448}/l$ up. Negative denominators and scalar division use the same endpoint rules. Each operation encloses all represented real values by monotonicity; induction proves inclusion for every expression built from them. Rectangular complex multiplication uses these real rules. The sum of coordinate halfwidths bounds the complex distance from its exact rational center. Binary floating arguments are never silently converted.

For $0 < x < 1$, the alternating arctangent sum and the endpoint obtained by adding the next signed term enclose $\arctan x$. Sixty terms at $1/5$ and $1/239$, together with Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, give the interval for π. The negative coefficient reverses the second interval. The identity follows from tangent addition and the range of the two angles; no decimal value of π is a trusted input.

For rational $x > 0$, range-reduce $x = 2^k v$, $1 \le v < 2$. With $z = (v-1)/(v+1) \le 1/3$,

$$\log v = 2\sum_{j=0}^{J-1} \frac{z^{2j+1}}{2j+1} + \mathcal{R}, \qquad 0 \le \mathcal{R} \le \frac{2z^{2J+1}}{(2J+1)(1-z^2)}.$$

The geometric tail bounds each remaining denominator below by $2J+1$. Use $J = 92$, the analogous expression for $\log 2$, and sign-aware interval multiplication by k. This covers negative k as well.

For complex z, repeatedly halve until its interval ℓ^1 magnitude bound is $A \le 1/2$. The degree 60 Taylor polynomial for e^z then has remainder at most

$$\frac{A^{61}}{61!}\frac{1}{1-A/62}.$$

Successive absolute terms have ratio at most $A/62$, so geometric comparison proves this bound. Add its symmetric interval to each coordinate, then square outward the required number of times. The identity $e^{2z} = (e^z)^2$ and interval inclusion prove the final enclosure. Caches are type-sensitive: otherwise a cached exact argument can accidentally bypass rejection of an equal-valued float. No implementation change is allowed during the recorded run; the producer verifies source bytes again before emitting its certificate.

4 Recomputing the compact growth constants

For the larger growth disk, $c = 1/2$, $r = 12/5$, take $k = 2$ when $a = 1$ and $k = 1$ when $a = 0$. Both real Mellin exponents on that disk are at most k. Define

$$Q_k(x) = \sum_{j=0}^{k} \frac{k!}{(k-j)!x^{j+1}}, \qquad U_{16}(x) = (1+x/16)^{-16}, \qquad \delta = 3/q.$$

Integration by parts gives $\int_1^\infty t^k e^{-xt}\,dt = e^{-x}Q_k(x)$. The same Gaussian split and geometric-tail comparison used below, now with cutoff $n_0 = 8$, imply the exact rational majorant

$$\widehat{M}_q = 2\left[\sum_{n=1}^{7} n^a U_{16}(\delta n^2)Q_k(\delta n^2) + C_a Q_k(64\delta)\frac{U_{16}(32\delta)}{1-U_{16}(8\delta)}\right], \quad C_0 = 1, \quad C_1 = (1+\delta^{-1})/2.$$

Replacing the actual $d = \pi/q$ by the smaller positive δ increases the integral majorant. Exact fraction comparison gives $\widehat{M}_3 < 4$, $\widehat{M}_4 < 10$, $\widehat{M}_5 < 6$. The producer recomputes these fractions, asserts the strict comparisons and exports them; the constants in the readout proof are thus checked rather than supplied by a floating growth estimate.

5 The completed-function integral and its tails

Let $d = \pi/q$ and $\theta_q(t) = \sum_{n\geq 1} \chi_q(n) n^a e^{-dn^2 t}$. The standard primitive theta transformation, followed by splitting the Mellin integral at1, gives

$$\Lambda_q(s) = \int_1^\infty \theta_q(t) t^{(s+a)/2-1}\, dt + \int_1^\infty \theta_q(t) t^{(1-s+a)/2-1}\, dt.$$

The second root-number coefficient is1 for these three characters: their finite Gauss sums are $i\sqrt{3}$, $2i$ and $\sqrt{5}$, respectively. For the last, $2\cos(2\pi/5) - 2\cos(4\pi/5) = \sqrt{5}$; the first two are direct conjugate-root differences. Division by $i^a\sqrt{q}$ gives the claimed normalization. Exponential decay makes the two integrals entire. This classical theta identity, not an assumption of Artin holomorphy or desired positivity, is the analytic input of the implementation.

The actual circle nodes are enclosed by computing $e^{2\pi i j/64}$ with the validated primitives. All input boxes fit $D = [-1, 2] + i[-4/3, 4/3]$. Thus both exponents $p = (s+a)/2-1$ and $p' = (1-s+a)/2-1$ satisfy $|\Re p| \leq 2$, $|\Im p| \leq 2/3$. The domain guard precedes all term discards. Retain $1 \leq n \leq 11$ and integrate $1 \leq t \leq 128$.

Put $\delta = 3/q < d$, $U(x) = (1+x/128)^{-128}$ and $Q(x) = x^{-1} + 2x^{-2} + 2x^{-3}$. Since $(1+x/128)^{128} \leq e^x$, $e^{-x} \leq U(x) < 1$ for $x > 0$. Integration by parts gives $\int_1^\infty t^2 e^{-xt}\, dt = e^{-x}Q(x)$. With $C_0 = 1$, $C_1 = (1+\delta^{-1})/2$ the omitted-n tail of both integrals is

$$2C_a Q(\delta 12^2) \frac{U(\delta 12^2/2)}{1 - U(12\delta)}.$$

Indeed $n^a e^{-\delta n^2/2} \leq C_a$, the function Q decreases, and $(12+j)^2 \geq 12^2 + 24j$. These inequalities yield a geometric bound after splitting the exponent in halves. The retained-n time tail is at most

$$\sum_{n=1}^{11} 2n^a 128^3 U(128\delta n^2) Q(128\delta n^2).$$

It follows by substituting $t = 128u$ and majorizing $t^{\Re p}$ by t^2. The two tails are disjoint: omitted n use the full integral, whereas retained n contribute only $t > 128$.

6 High-order integration with a certified remainder

Split each dyadic interval $[2^j, 2^{j+1}]$, $j = 0, \ldots, 6$, into four equal panels. For its midpoint m and halfwidth h, $h/m \leq 1/9$. For $f(z) = e^{-\lambda z} z^p$, $\lambda = \pi n^2/q$, the principal logarithm is analytic on $|z - m| \leq \rho = m/2$. On that circle,

$$|f(z)| \leq M_\rho = 3\max\{(3m/2)^2, (m/2)^{-2}, 1\} U(\delta n^2 m/2).$$

Here $\Re z \geq m/2$, $|\arg z| \leq 1$ and $e^{|\Im p|} \leq e^{2/3} \leq 3$. The maximum handles both signs of $\Re p$. Cauchy's coefficient estimate and cancellation of odd powers on $[-h, h]$ therefore bound the error after degree $2K$ by

$$\frac{2hM_\rho (h/\rho)^{2K+2}}{(2K+3)(1-(h/\rho)^2)}.$$

Indeed integrate each remaining even term, replace its denominator by the first omitted denominator, then sum a geometric series. Here $K = 12$. Multiply by $2n^a$ for both Mellin terms and sum all 28 panels and retained n. Keeping this majorant even for later discarded terms only overcounts error.

To evaluate the polynomial stably, set $u = h/m$, $q_{-1} = 0$, $q_0 = 1$ and use

$$q_{j+1} = \frac{u(p - \lambda m - j)q_j - (\lambda m)u^2 q_{j-1}}{j+1}.$$

Multiplying $f' = (-\lambda + p/t)f$ by $t = m + x$ and matching coefficients proves this recurrence, with $q_j = f^{(j)}(m)h^j/(j!f(m))$. The integrated polynomial is $2he^{-\lambda m}m^p \sum_{j=0}^{K} q_{2j}/(2j+1)$; all its arithmetic is outward.

Fixed absolute exponential-rounding errors can be amplified by high-order coefficients in late panels. The code therefore first bounds a complete (n, panel) pair, including both powers, by

$$4hn^a(m+h)^2 U(\delta n^2(m-h)).$$

For real $t \geq 1$ its phase has modulus1 and $t^{\Re p} \leq (m+h)^2$, proving the bound. If it is below 2^{-90}, skip that pair entirely and add its bound to the final radius. This is controlled truncation, not numerical underflow. The final sample radius is the interval-sum halfwidth plus the omitted-n, time, Cauchy and discard bounds. All are recorded separately and checked as nonnegative. No high-precision special-function routine is trusted here.

7 Reproduction and remaining interfaces

Run the preserved source on the existing Linux Python environment with no new packages: `python3` on `examples/analytic_oracle/theta_oracle.py`, with argument `-output NEW`. It refuses to overwrite an existing experiment directory. The final-v5 experiment contains 192 samples, three rational degree 28 polynomials, copied source and SHA-256 hashes. Independent validation checks rational endpoint arithmetic, transcendental enclosures, 1160 exact coefficient identities, high- precision completed-function comparisons and the exported component sums.

The fixed-parameter oracle solves a concrete evaluation/certification gap. It does not identify a positive arithmetic operator, force real-rooted polynomials, prove the signed original-family short-row bound or supply complete Lean4 verification. Surface/order and neural architectures may use these certificates as independent targets, but a small learned loss is not a substitute for the remaining arithmetic and categorical theorems.

8 Requested-error readouts from preserved certified labels

The accepted final-v5 sample certificate is preserved unchanged. For each $q = 3, 4, 5$, it supplies $m = 64$ rational complex labels y_j, bounds $|y_j - \Lambda_q(c + \rho\zeta_j)| \leq \eta_j$, and outward rectangles enclosing the true roots $\zeta_j = e^{2\pi ij/m}$, where $c = 1/2$, $\rho = 6/5$. The classical primitive theta/Mellin identity and accepted growth bounds $M_q = 4, 10, 6$ on $|s - c| \leq r = 12/5$ remain the analytic inputs. This extension accepts an exact rational error request $\varepsilon > 0$ on the fixed compact $|s - c| \leq R = 7/10$. It changes the finite readout, while retaining the certified sample errors and the 64-node alias budget. It is not an arbitrary-accuracy theta-sampling interface.

Theorem 2 (Executed rational readout with explicit refusal)**.** *For each supported conductor and rational request, the finite-data selection below either returns a normalized rational polynomial P*

and an executed, exactly replayable gradient-descent trace with

$$\sup_{|s-c|\le R} |P(s) - \Lambda_q(s)| \le \varepsilon,$$

or refuses the request with the exact minimum base budget over all 64 *allowed degrees. Refusal concerns attainability by this particular fixed-data majorant; it is not an impossibility theorem for approximation.*

Proof. Let B_n be an outward rectangle enclosing $b_n^* = m^{-1}\sum_j y_j \bar{\zeta}_j^n$, $0 \le n < 64$, computed from the preserved root rectangles. Put $t_n = \mathrm{mid}(B_n)$ and let e_n be the sum of its coordinate halfwidths. These are exact rationals, and $|t_n - b_n^*| \le e_n$. For a selected $d < 64$ execute, using exact rational operations,

$$b_n^0 = 0, \qquad b_n^{k+1} = \tfrac{3}{4} b_n^k + \tfrac{1}{4} t_n, \qquad 0 \le n \le d.$$

This is genuine gradient descent with step $1/4$ for the explicitly specified real Euclidean objective $J(b) = \frac{1}{2}\sum_{n=0}^{d} |b_n - t_n|^2$, whose gradient is $b - t$. All iterates and exact objective values are retained. The output is literally

$$P_k(s) = \sum_{n=0}^{d} b_n^k \left(\frac{s-c}{\rho}\right)^n.$$

Set $u = R/\rho = 7/12$, $v = R/r = 7/24$, $w = (\rho/r)^{64} = 2^{-64}$, $S_d(x) = \sum_{n=0}^{d} x^n$ and $\eta = \max_j \eta_j$. The five budgets are

$$E_{\mathrm{opt}}(k) = \sum_{n=0}^{d} |b_n^k - t_n|_1 u^n, \qquad E_{\mathrm{oracle}} = \eta S_d(u),$$

$$E_{\mathrm{alias}} = M_q \frac{w}{1-w} S_d(v), \qquad E_{\mathrm{Taylor}} = M_q \frac{v^{d+1}}{1-v},$$

$$E_{\mathrm{arithmetic}} = \sum_{n=0}^{d} e_n u^n.$$

Here $|x + iy|_1 = |x| + |y|$. Cauchy's estimate and true-root Fourier orthogonality give the alias and Taylor terms; certified label errors give the oracle term. Triangle inequality applied to $b_n^k - b_n^* = (b_n^k - t_n) + (t_n - b_n^*)$ gives the other two. Thus their exact sum bounds the compact-uniform error.

Define $C_d = E_{\mathrm{oracle}} + E_{\mathrm{alias}} + E_{\mathrm{Taylor}} + E_{\mathrm{arithmetic}}$. Select the smallest $d < 64$ with $C_d < \varepsilon$; if none exists, report $\min_{d<64} C_d$ and refuse. For a selected degree, $E_{\mathrm{opt}}(k) = (3/4)^k \sum_n |t_n|_1 u^n \longrightarrow 0$. Therefore executing updates until $E_{\mathrm{opt}}(k) \le \varepsilon - C_d$ terminates after finitely many steps. The serialized coefficients, budgets, targets and trace use exact rational endpoints; displayed decimal approximations are not proof inputs. □

9 Objective semantics, precision floors and replay

The original true-circle sample-label least-squares objective is

$$\mathcal{L}_y(b) = \frac{1}{2m} \sum_{j=0}^{m-1} \left| \sum_{n=0}^{d} b_n \zeta_j^n - y_j \right|^2.$$

With $d < m$, Fourier orthogonality gives $A^*A = mI$ and the real Euclidean coordinate gradient $\nabla \mathcal{L}_y(b) = b - b^*$. The executed objective J instead has the exact rational midpoint target t. The

difference $\nabla\mathcal{L}_y - \nabla J = t - b^*$ is enclosed coefficientwise by e_n. Consequently this is an actual rational GD run for J and a certified perturbed-gradient implementation for $\mathcal{L}_y$. It is not the exact optimizer trajectory of the ideal-root objective, nor a replay of neural or quantum training. An interval enclosure of an ideal-root GD trajectory would be a different explicitly labelled object.

Each actual update contracts $b^k - t$ by $3/4$, so the exact stored losses satisfy $J(b^{k+1}) = (9/16)J(b^k)$. A replay recomputes every rational coordinate update, loss and weighted residual. The closed form $b^k = (1 - (3/4)^k)t$ is an independent check on the executed trace, not its producer. Exact rational loop coordinates introduce no further midpoint or floating export error. The arithmetic budget covers the difference between the certified Fourier rectangles and their midpoint targets, including the inherited root enclosure and outward interval operations. A subsequent float conversion requires its own exact conversion-error bound.

The normalized polynomial convention is essential: a stored coefficient with index n multiplies $((s - 1/2)/(6/5))^n$, not $(s - 1/2)^n$. The input certificate is pinned by SHA-256, including its source hashes, all 192 sample enclosures, root rectangles and analytic radius components. Producing source is captured at entry, checked unchanged before emission, and copied into a fresh output directory. An existing output is never overwritten. An independent replay should also check the fixed input hash, all dimensions, degree choice, the first stopping iterate, all five exact terms and their literal sum against the requested rational tolerance.

The attainable boundary of this majorant is

$$C_{\min,q} = \min_{0 \le d < 64} C_{q,d}.$$

Every request $\varepsilon > C_{\min,q}$ has strict budget slack for some degree and the proved finite stop rule. At or below that boundary this interface refuses. The degree minimizing C_d need not be 63: Taylor error decreases with degree while the oracle and arithmetic sums increase. In particular $C_d \ge \eta$ and $C_d \ge M_q 2^{-64}/(1-2^{-64})$. Increasing the number of optimization steps does not lower these fixed-data budgets. Approximate boundary values on the stated compact are $7.2929 \cdot 10^{-17}$, $1.7370 \cdot 10^{-16}$ and $3.3741 \cdot 10^{-16}$ for $q = 3, 4, 5$; only the exact exported rational minima are used for acceptance. These are certificate floors, not lower bounds on the actual errors.

A full adaptive theta oracle would separately need proof-preserving choices of omitted-n cutoff, time cutoff, quadrature order, whole-panel discard budget, primitive series lengths, interval precision and serialized-bound precision. The present $N = 11$, $T = 128$, 224-bit arithmetic, 60/92/60 primitive series lengths, 2^{-90} discard threshold and 180-bit rounded budgets cannot support every positive request. Increasing a nominal bit count alone leaves the fixed primitive remainders and analytic tails in place. The preserved labels are not recomputed here. The classical identity, finite arithmetic and informal proof are not new Lean declarations. No zero-location, all-conductor, positive arithmetic determinant, quantum-advantage or GRH conclusion follows.

The interactive replay is available at the certified readout laboratory. All decimal graphics are illustrative views of the linked exact records.

10 An arithmetic-family moment with every original row retained

Let $K = \mathbb{Q}(\sqrt{-3})$, $\mathcal{O} = \mathbb{Z}[\omega]$, and write N for the ideal or element norm. Fix a finite-order Hecke character ν and a finite set S of prime ideals containing those above 6 and those dividing its conductor. For $0 \ne u \in \mathcal{O}$, let κ_u be the character of $K(u^{1/6})/K$ defined using arithmetic Frobenius, and put $\psi_u = \mathrm{prim}(\nu\kappa_u)$. Here prim means the underlying primitive Hecke character: it is formed before introducing any divisor variable or evaluating conductor-dependent zero extensions. The rows below are all actual Eisenstein elements, including units and prime powers supported in S.

The external inputs are abelian Kummer reciprocity and character uniqueness, the primitive Hecke functional equation and its usual vertical growth, Stirling's formula, scalar Dedekind-zeta convexity, and Blomer–Goldmakher–Louvel's large sieve for their specified sixth-order family. We use the pinned arXiv:1112.1650v1, Theorem 1.3, Lemma 2.1 and Appendix II; its Section 5 supplies the all-ideal decomposition whose fixed-twist proof is redone here. No statement about all primitive characters of comparable conductor, Artin holomorphy, Lindelöf bounds or zero locations is assumed.

Theorem 3 (The original-family numerator moment)**.** *For every $\varepsilon > 0$, uniformly for $H \geq 1$ and $t \in \mathbb{R}$,*

$$\sum_{0<Nu\leq H} |L(1/2+it,\psi_u)|^2 \ll_{\nu,S,\varepsilon} H^{1+\varepsilon}(1+|t|)^{1+\varepsilon}. \tag{1}$$

For every fixed $0 < \delta < 1/2$, the same estimate holds with $1/2+it$ replaced by $1/2+\delta+it$, with the constant also depending on δ. If the central bound has constant C_0 at an exponent $0 < \varepsilon_0 \leq 1/4$, an admissible off-central constant is

$$200C_0\left(\frac{3/2+\delta}{1/2-\delta}\right)^2. \tag{2}$$

Proof. For the canonical BGL character $\vartheta_{\mathfrak{a}}$, Appendix II gives a radicand $\alpha_{\mathfrak{a}}$ with $(\alpha_{\mathfrak{a}})_S = \mathfrak{a}\mathfrak{g}^{-6}$. Since $\mathcal{O}_S$ is principal, comparison with any literal u whose outside-S ideal is $\mathfrak{a}$ gives $\kappa_u = \eta\vartheta_{\mathfrak{a}}$ for one of finitely many fixed S-unit Kummer twists η. This is an equality of primitive characters, established outside a finite set and extended by uniqueness. For each outside-S ideal $\mathfrak{a}$ and S-supported integral ideal $\mathfrak{b}$, at most six elements generate $\mathfrak{a}\mathfrak{b}$. The resulting S-supported multiplicity sum converges at exponent $1+\varepsilon$.

For each fixed $\xi = \nu\eta$, split the canonical parameter as $\mathfrak{a} = \mathfrak{a}_0\mathfrak{a}_1\mathfrak{a}_2^6$, where $\mathfrak{a}_0$ is squarefree, $\mathfrak{a}_1$ is squarefull and sixth-power-free, and $(\mathfrak{a}_0, \mathfrak{a}_1) = 1$. The primitive conductor is at most a fixed multiple of $N(\mathfrak{a}_0\mathfrak{a}_1)$. The common-gamma approximate functional equation, with $X = H/(N\mathfrak{a}_2)^6$ and $T = 1+|t|$, has common maximal length $B = X^{1/2}T(HT)^{\eta}$ for any sufficiently small fixed $\eta > 0$. Mellin inversion removes its row-dependent conductor weight. Split a column as $\mathfrak{b}_S\mathfrak{b}_1\mathfrak{b}_2$, with $\mathfrak{b}_S$ supported in S, $\mathfrak{b}_1$ squarefree and $\mathfrak{b}_2$ squarefull, the last two coprime and prime to S. The original primitive values at $\mathfrak{b}_S$ remain bounded row multipliers. The S-supported and squarefull column weights are summable at $1/2+\eta$. Multiplying squarefree column coefficients by the fixed ξ preserves their squared norm, whatever its finite order. BGL's large sieve therefore bounds the energy for fixed $\mathfrak{a}_1, \mathfrak{a}_2$ by

$$(HT)^{O(\eta)}\{M + B + (MB)^{2/3}\}, \qquad M = X/N\mathfrak{a}_1.$$

Squarefull counting and summation give $X + B\sqrt{X} + (XB)^{2/3} \ll (HT)^{O(\eta)}XT$. The $\mathfrak{a}_2$ sum converges at exponent 6. Principal rows are separated before the pole-free functional equation and bounded by scalar zeta convexity. This proves (1).

For the off-central statement divide each $L(s,\psi)$ by $(s+1)^{5/4}$ and, on principal rows, also multiply by $(s-1)/(s+1)$. These are bounded analytic functions on $\operatorname{Re} s > 1/2$, continuous on the boundary. Poisson representation and Jensen's inequality transfer the central squared moment. The convolution of $P_\delta(y) = \delta/[\pi(\delta^2+y^2)]$ with $(1+|t|)^{-3/2+\varepsilon_0}$ is at most eleven times that weight when $0 < \varepsilon_0 \leq 1/4$. Restoring the denominator and the principal pole-removal factor gives (2). The extended edition gives the detailed derivation with all primitive local values. □

Separate fixed-unitary extension. The moment theorem also holds for any one fixed unitary Hecke character ν. For precision choose $\nu(\mathfrak{a}) = \nu_0(\mathfrak{a})(N\mathfrak{a})^{-i\tau}$, with angular parameter $m \in \mathbb{Z}$ and

completed gamma factor

$$\gamma_\nu(s) = 2(2\pi)^{-(s+|m|/2+i\tau)}\Gamma(s+|m|/2+i\tau).$$

All finite Kummer twists have this same infinity type. Use $T_\nu = 1+|t+\tau|+|m|$, comparable to $1+|t|$ with ν-dependent constants. Pure norm rows have $L(s,\psi) = \zeta_K(s+i\tau)$ and are separated before the pole-free functional equation; their pole at $1-i\tau$ is removed by $(s-1+i\tau)/(s+1+i\tau)$. No uniformity in m, τ or the conductor of ν is asserted. This extension of the moment is separate from the following finite-order type-I residue formula.

10.1 The exact masked remainder and the remaining loss

Retain the original zero-extended symbol $\chi_d(u) = (u/d)_6$ on good squarefree ideals. Let μ be the ideal Möbius function and set

$$\beta_U(d) = \sum_{\substack{ab=d\\ Na,Nb\le U}} \mu(a)\mu(b), \quad T(u,d) = S \cup \operatorname{supp}(u) \cup \operatorname{supp}(d).$$

Let $\psi_{u,2} = \operatorname{prim}(\psi_u^2)$, formed independently of squaring any zero sentinel. With $\mathfrak{f}_u$ and $\mathfrak{f}_{u,2}$ the primitive conductors, the full-mask series is

$$F_{u,d}(s) = \frac{L(s,\psi_u)}{L(2s,\psi_{u,2})} C_{T(u,d)}(s), \tag{3}$$
$$C_T(s) = \frac{\prod_{p\in T,\ p\nmid \mathfrak{f}_u}(1-\psi_u(p)(Np)^{-s})}{\prod_{p\in T,\ p\nmid \mathfrak{f}_{u,2}}(1-\psi_{u,2}(p)(Np)^{-2s})}.$$

Initially it equals $\sum_{c \text{ squarefree},\ (c,T)=1} \psi_u(c)(Nc)^{-s}$ for $\operatorname{Re} s > 1$. Every prime of T remains deleted, including those forgotten by either primitive conductor.

For finite-order ν, $D \ge 1$, $U \ge 1$, $U^2 \le D$, and $W \in C_c^\infty((0,\infty))$ supported in $[1,2]$, put

$$\widehat{W}(s) = \int_0^\infty W(x)x^{s-1}\,dx, \qquad \mathcal{M}_4(W) = \sum_{j=0}^{4} \sup_x |(x\partial_x)^j W(x)|.$$

On $\sigma = 1/2+\delta$, the exact residue-subtracted type-I remainder is

$$E_u = \frac{D^\delta}{2\pi}\int_{\mathbb{R}} \widehat{W}(\sigma+it)D^{it} \sum_{\substack{d \text{ squarefree}\\ (d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{(Nd)^{\sigma+it}} F_{u,d}(\sigma+it)\,dt. \tag{4}$$

This definition retains the actual β_U, all phases, d-masks, conductors and the independently primitive square.

Theorem 4 (An improved masked residual estimate)**.** *Under the preceding finite-order hypotheses, for every $\varepsilon > 0$,*

$$\sum_{0<Nu\le H} |E_u|^2 \ll_{\nu,S,\delta,\varepsilon} \mathcal{M}_4(W)^2 D^{2\delta} U^{2-4\delta} H(HD)^\varepsilon. \tag{5}$$

Proof. The same deleted set in both factors of (3) gives $|F_{u,d}(\sigma+it)| \ll (Nu\,Nd)^\eta |L(\sigma+it,\psi_u)|$: the reciprocal incomplete Euler product at $2\sigma > 1$ is bounded by $\zeta_K(2\sigma)$, and its remaining finite

numerator is bounded by a small power of its radical. Apply Minkowski to (4) in the exact row space and use Theorem 3. The actual coefficient sum is at most

$$\sum_d \frac{|\beta_U(d)|}{(Nd)^{\sigma-\eta}} \le \left(\sum_{Na \le U} (Na)^{-\sigma+\eta} \right)^2 \ll U^{1-2\delta+2\eta}.$$

Four Mellin integrations by parts bound the remaining height integral by a constant times $\mathcal{M}_4(W)$. Square the result and absorb the small losses using $U^2 \le D$. □

At $H = D^{3/4}$ and $U = D^a$ with fixed $0 < a \le 1/2$, the upper bound's excess over H is $2\delta + (2 - 4\delta)a$, up to small losses; as $\delta \downarrow 0$ it still leaves $2a$. This is a limitation of the upper bound, not an energy lower bound. The extracted principal residue, the original short term at smaller scales and the signed type-II minus type-I covariance are not controlled by (5). The next arithmetic task is to estimate the complete d-weighted moment before its absolute coefficient sum, or establish sufficient signed covariance. No positive arithmetic determinant, zero-location theorem or GRH proof follows. The analytic argument uses the external classical inputs above; finite diagnostics and selected existing Lean local identities do not formalize the Hecke functions, approximate functional equation, Poisson transfer or these moment bounds.

11 Family identification, the approximate functional equation and masks

We prove the analytic statements of Section 10 from the explicit classical inputs stated there. Write $I(S)$ for the nonzero integral ideals prime to S, and $\mathcal{O}_S$ for the ring of S-integers. Constants may depend on the fixed field, S, ν and any declared positive small exponent. No such constant is asserted uniform in the conductor or infinity type of a varying ν.

11.1 A finite comparison with the literal Kummer rows

Use Appendix II of BGL, with $n = 6$. Its fixed local modulus and representatives may be chosen once, since S contains the primes above 6 and $\mathcal{O}_S$ has class number one. For each $\mathfrak{a} \in I(S)$ the construction writes

$$\mathfrak{a} = (x)E\mathfrak{g}^6, \qquad \alpha_{\mathfrak{a}} = xm_E, \qquad (m_E)_S = E,$$

so $(\alpha_{\mathfrak{a}})_S = \mathfrak{a}\mathfrak{g}^{-6}$. Choose $y \in K^\times$ generating $\mathfrak{g}$ as an $\mathcal{O}_S$-ideal. If $(u)_S = \mathfrak{a}$, then

$$e = \alpha_{\mathfrak{a}} y^6/u \in \mathcal{O}_S^\times, \qquad [\alpha_{\mathfrak{a}}] = [eu] \quad \text{in } K^\times/K^{\times 6}.$$

There is a precision here: Appendix II allows choosing $\mathfrak{g}$ relative to a tested denominator. Fix one admissible global decomposition first. Its radicand gives the canonical character $\vartheta_{\mathfrak{a}}$ on all primes outside the finite set $S \cup \mathrm{supp}(\mathfrak{a}\mathfrak{g})$ where that choice applies. Uniqueness of Hecke characters, equivalently Chebotarev in the finite abelian quotient, then identifies the underlying primitive characters everywhere. With matching arithmetic-Frobenius conventions,

$$\kappa_u = \kappa_{e^{-1}} \vartheta_{\mathfrak{a}}. \tag{6}$$

If the residue-symbol convention is conjugated, conjugate the canonical family and its coefficient sequence; the same large sieve follows.

The group $\mathcal{O}_S^\times/(\mathcal{O}_S^\times)^6$ is finite. Thus (6) uses one of a finite set of fixed Kummer twists η, all unramified outside S and of bounded S-conductor. This is a comparison of primitive characters, not a pointwise multiplication rule for independently zero-extended arrays at bad primes. In particular, a conductor drop must not be replaced by a product of artificial zeros.

Since $\mathcal{O}$ is principal with six units, for fixed $\mathfrak{a} \in I(S)$ and an S-supported integral ideal $\mathfrak{b}$ there are at most six u satisfying $(u) = \mathfrak{a}\mathfrak{b}$. For any nonnegative row quantity, the literal moment is consequently at most

$$6 \sum_{\substack{\mathfrak{b}\ S\text{-supported} \\ N\mathfrak{b} \le H}} \sum_{\eta} \sum_{\substack{\mathfrak{a} \in I(S) \\ N\mathfrak{a} \le H/N\mathfrak{b}}} \left| L(s, \mathrm{prim}(\nu\eta\vartheta_{\mathfrak{a}})) \right|^2 . \tag{7}$$

An ideal bound $X^{1+\varepsilon}T^{1+\varepsilon}$ therefore transfers to all literal rows, since

$$\sum_{\mathfrak{b}\ S\text{-supported}} (N\mathfrak{b})^{-1-\varepsilon} = \prod_{p \in S} (1 - (Np)^{-1-\varepsilon})^{-1} < \infty.$$

No quotient of the original unit rows or deletion of S-prime powers has occurred.

11.2 Fixed twists, exact tame conductors and the squarefree sieve

Fix $\xi = \nu\eta$ and write $\rho_{\mathfrak{a}} = \mathrm{prim}(\xi\vartheta_{\mathfrak{a}})$. Its fixed conductor is supported in S before the Kummer twist. On columns prime to S, $|\xi(\mathfrak{b})| = 1$. BGL Theorem 1.3 thus gives, for any fixed $\mathfrak{a}_1$,

$$\begin{aligned} &\sum_{\substack{\mathfrak{a}_0 \in I(S)\ \text{squarefree} \\ N\mathfrak{a}_0 \le M}} \left| \sum_{\substack{\mathfrak{b}_1 \in I(S)\ \text{squarefree} \\ N\mathfrak{b}_1 \le B}} \lambda_{\mathfrak{b}_1} \xi(\mathfrak{b}_1) \vartheta_{\mathfrak{a}_1}(\mathfrak{b}_1) \vartheta_{\mathfrak{a}_0}(\mathfrak{b}_1) \right|^2 \\ &\qquad \ll (MB)^{\varepsilon} \{M + B + (MB)^{2/3}\} \sum_{\mathfrak{b}_1} |\lambda_{\mathfrak{b}_1}|^2. \end{aligned} \tag{8}$$

The first fixed multiplier preserves the coefficient norm; the second has modulus at most one. Restricting the positive outer row sum is harmless. A coprimality condition with another fixed ideal belongs to λ. In particular, ν need not have order dividing six and is not asserted to belong to BGL's finite ray-class character set.

There is a unique valuation decomposition

$$\mathfrak{a} = \mathfrak{a}_0\mathfrak{a}_1\mathfrak{a}_2^6, \quad \mathfrak{a}_0 \text{ squarefree}, \quad \mathfrak{a}_1 \text{ squarefull and sixth-power-free}, \quad (\mathfrak{a}_0, \mathfrak{a}_1) = 1.$$

Indeed, for $v_p(\mathfrak{a}) = 6q + r$, $0 \le r \le 5$, put the residual prime in $\mathfrak{a}_0$ if $r = 1$, and its exponent r in $\mathfrak{a}_1$ if $2 \le r \le 5$. BGL Lemma 2.1 gives $\vartheta_{\mathfrak{a}\mathfrak{b}^6} = \vartheta_{\mathfrak{a}}$ and $\vartheta_{\mathfrak{a}_0\mathfrak{a}_1} = \vartheta_{\mathfrak{a}_0}\vartheta_{\mathfrak{a}_1}$ on ideals prime to S in this sixth-power-free case. Those are the identities needed here; no arbitrary multiplicativity assertion is substituted.

At a prime outside S, tame Kummer inertia at residual exponent r has order $6/\gcd(6, r)$. It is nontrivial for $r = 1, \dots, 5$ and has conductor exponent one. The fixed ξ is unramified there, so cannot cancel that inertia. At primes in S the local sixth-power classes are finite, hence only finitely many primitive conductor types occur after multiplication by the fixed ξ. Consequently

$$Q_{\rho_{\mathfrak{a}}} = |D_K| N\mathfrak{f}_{\rho_{\mathfrak{a}}} \le C_{\xi,S} N(\mathfrak{a}_0\mathfrak{a}_1), \tag{9}$$

independently of $\mathfrak{a}_2$. The equality of the conductor's outside-S part with $\mathrm{rad}(\mathfrak{a}_0\mathfrak{a}_1)$ does not identify that conductor with the original larger deleted set.

11.3 Removing the row-dependent functional-equation weights

First take finite-order ν. All characters then have trivial infinity type at the complex place. The primitive completion may be normalized as

$$\Lambda(s,\rho) = Q_\rho^{s/2}\gamma_K(s)L(s,\rho), \qquad \gamma_K(s) = 2(2\pi)^{-s}\Gamma(s), \qquad \Lambda(s,\rho) = \epsilon_\rho\Lambda(1-s,\bar\rho), \quad |\epsilon_\rho| = 1.$$

For nonprincipal rows the completed function is entire. Apply this functional equation to the Mellin integral with even auxiliary factor $G(z) = e^{z^2}$ and denominator z, shifting across $z = 0$. At $s = 1/2 + it$ this gives two weighted Dirichlet series, truncated below to polynomials; the dual prefactor has modulus one. Principal rows are excluded from this pole-free step and treated separately below.

For fixed $\mathfrak{a}_2$, let $X = H/(N\mathfrak{a}_2)^6 \geq 1$ and $T = 1 + |t|$. An explicit normalized kernel is

$$V_t(x) = \frac{1}{2\pi i}\int_{(\eta)} e^{z^2}\frac{\Gamma(s+z)}{\Gamma(s)}T^{-z}x^{-z}\,\frac{dz}{z}, \qquad x = \frac{2\pi N\mathfrak{b}}{\sqrt{Q_\rho}T}, \tag{10}$$

with $\eta > 0$ small. Its factors combine to $(\sqrt{Q_\rho}/(2\pi N\mathfrak{b}))^z$ in the actual completed-L kernel; there is no extra factor of T. The denominator z supplies the residue at zero. On $z = \eta + iy$, Stirling and $|t + y| \leq T(1 + |y|)$ give

$$\left|\frac{\Gamma(1/2+\eta+i(t+y))}{\Gamma(1/2+it)T^{\eta+iy}}\right| \ll_\eta (1+|y|)^{C_\eta}e^{\pi|y|/2}.$$

The factor e^{-y^2} makes this an integrable majorant uniformly in t. The analogous bound holds on every fixed positive shifted line and for the dual gamma kernel. Moving far enough to the right gives arbitrary power decay beyond a common total-ideal cutoff

$$B' = X^{1/2}T(HT)^\eta, \tag{11}$$

up to fixed ξ, S, K constants. For any requested $A > 0$, a sufficiently large positive kernel shift, depending also on η, makes the truncation error $O((HT)^{-A})$ uniformly in all rows in this block. Ideal counting $O_K(Y)$ bounds the absolute tails. Such errors are negligible after the finite row sums by taking A larger if needed.

Insert (10) on $\operatorname{Re} z = \eta$. The factor $(\sqrt{Q_\rho}T)^z$ is independent of the column and is a row multiplier of modulus at most $(CXT^2)^{\eta/2}$. Its imaginary part is a phase. The remaining column coefficient is $(N\mathfrak{b})^{-1/2-\eta-it-iy}$, apart from fixed constants. Minkowski or Cauchy with the integrable gamma majorant therefore removes the row-dependent weight at cost $(HT)^{O(\eta)}$ in the squared norm. No independence of the root numbers is assumed or needed.

For each column use the unique decomposition

$$\mathfrak{b} = \mathfrak{b}_S\mathfrak{b}_1\mathfrak{b}_2, \quad \mathfrak{b}_S\ S\text{-supported}, \quad \mathfrak{b}_1\ \text{squarefree}, \quad \mathfrak{b}_2\ \text{squarefull}, \quad (\mathfrak{b}_1,\mathfrak{b}_2) = 1,$$

where $\mathfrak{b}_1, \mathfrak{b}_2 \in I(S)$; a prime of outside-$S$ valuation one goes in $\mathfrak{b}_1$, and a prime of valuation at least two goes with its full exponent in $\mathfrak{b}_2$. For fixed $\mathfrak{b}_S, \mathfrak{b}_2$ retain $\rho_{\mathfrak{a}}(\mathfrak{b}_S)$ as the original primitive row value. On the good factors the exact canonical identity applies. The remaining factors involving $\mathfrak{b}_2$, including $\vartheta_{\mathfrak{a}_0}(\mathfrak{b}_2)$, are bounded row multipliers. Their moduli are at most one; only then are they discarded in an inequality. In particular the value at $\mathfrak{b}_S$ is not replaced by the product of zero-extended primitive arrays.

For the $\mathfrak{b}_1$ polynomial the coefficient sequence is independent of $\mathfrak{a}_0$; it includes the indicator $(\mathfrak{b}_1, \mathfrak{b}_2) = 1$ and the common cutoff $N\mathfrak{b}_1 \le B'/N(\mathfrak{b}_S\mathfrak{b}_2)$. Padding it by zeros up to B' is legitimate. Its squared norm is at most $\zeta_K(1+2\eta)$. Minkowski over $\mathfrak{b}_S, \mathfrak{b}_2$ is legitimate because

$$\sum_{\mathfrak{b}_S,\mathfrak{b}_2} N(\mathfrak{b}_S\mathfrak{b}_2)^{-1/2-\eta} < \infty.$$

The finite Euler product handles $\mathfrak{b}_S$; the squarefull series converges above $1/2$. Apply (8) to each such polynomial and to the conjugate dual family. For fixed $\mathfrak{a}_1, \mathfrak{a}_2$ the resulting energy is

$$(HT)^{O(\eta)}\{M + B' + (MB')^{2/3}\}, \qquad M = X/N\mathfrak{a}_1. \tag{12}$$

The additional condition $(\mathfrak{a}_0, \mathfrak{a}_1) = 1$ simply restricts the positive row sum. The large-sieve small exponent can be chosen sufficiently small along with η and absorbed in the indicated small loss.

11.4 All-ideal summation and principal rows

Every squarefull ideal $\mathfrak{c}$ has a unique expression $\mathfrak{d}^2\mathfrak{e}^3$ with $\mathfrak{e}$ squarefree. This follows primewise by using exponent zero or one in $\mathfrak{e}$ according to the parity of an exponent at least two. Therefore

$$\#\{\mathfrak{c} \text{ squarefull} : N\mathfrak{c} \le X\} \ll_K X^{1/2}, \qquad \sum_{\mathfrak{c} \text{ squarefull}} (N\mathfrak{c})^{-s} < \infty \quad (s > 1/2).$$

For the count, sum $O((X/(N\mathfrak{e})^3)^{1/2})$ and use $\zeta_K(3/2) < \infty$. Summing (12) over $\mathfrak{a}_1$ gives

$$(HT)^{O(\eta)}\{X + B'\sqrt{X} + (XB')^{2/3}\} \ll (HT)^{O(\eta)}XT,$$

since the squarefull series converges at 1 and $2/3$, $B' = \sqrt{X}\,T(HT)^{\eta}$, and $T^{2/3} \le T$ for $T \ge 1$. The $\mathfrak{a}_2$ sum is bounded by $H\sum_{\mathfrak{a}_2}(N\mathfrak{a}_2)^{-6} = H\zeta_K(6)$. Choosing all internal losses small for the requested final ε proves the fixed-twist all-ideal central moment.

There are $O_K(H)$ ideal rows in total. On any row where the primitive character is principal its L-function is ζ_K, and ordinary scalar convexity gives $|\zeta_K(1/2+it)|^2 \ll_\varepsilon T^{1+\varepsilon}$. This treats those rows without a pole-free approximate functional equation. Finally (7) proves (1) for every actual nonzero Eisenstein element.

11.5 Poisson transfer with an explicit constant

For a nonprincipal row define

$$f_\psi(s) = \frac{L(s,\psi)}{(s+1)^{5/4}},$$

using the holomorphic logarithm on $\mathrm{Re}(s+1) > 0$. On a principal row additionally multiply by $(s-1)/(s+1)$. Analytic continuation and ordinary degree-two convexity, used separately at each fixed conductor, show that f_ψ is bounded analytic on the right half-plane $\mathrm{Re}\, s > 1/2$, with continuous boundary and decay at infinity. The principal pole is removed before applying Poisson representation to its real and imaginary parts. Jensen gives

$$|f_\psi(1/2+\delta+it)|^2 \le \int_{\mathbb{R}} P_\delta(t-v)|f_\psi(1/2+iv)|^2\,dv.$$

The exact row family is finite for fixed H, so its sum interchanges with the integral without a limiting-family issue. The boundary pole-removal factor has modulus at most one.

Take $0 < \varepsilon_0 \le 1/4$ and $\alpha = 3/2 - \varepsilon_0 \in [5/4, 3/2)$. Since $|3/2 + iv| \ge (1 + |v|)/\sqrt{2}$, the central squared moment of the f_ψ is bounded by $3C_0 H^{1+\varepsilon_0} w_\alpha(v)$, where $w_\alpha(v) = (1 + |v|)^{-\alpha}$. For $|t| \ge 2$, split $|v - t| \le |t|/2$ from its complement. The near contribution to $P_\delta * w_\alpha$ is at most $2^\alpha w_\alpha(t) < 3w_\alpha(t)$. In the far region use $P_\delta(t - v) \le 4\delta/(\pi t^2)$ and $\int w_\alpha = 2/(\alpha - 1)$, giving at most

$$\frac{2\delta 3^\alpha}{\pi(\alpha - 1)} w_\alpha(t) \le 8 w_\alpha(t).$$

Here $(1 + |t|)^\alpha/t^2 \le 3^\alpha/4$ for $|t| \ge 2$. For $|t| \le 2$, the convolution is at most one and hence at most $3^\alpha w_\alpha(t) < 6w_\alpha(t)$. Thus $P_\delta * w_\alpha \le 11 w_\alpha$ uniformly for $0 < \delta < 1/2$. Restoring the denominator costs less than $6(1 + |t|)^{5/2}$, and the inverse principal factor is bounded by $(3/2 + \delta)/(1/2 - \delta)$. The product $11 \cdot 3 \cdot 6 = 198 < 200$ proves (2). The constant is uniform as $\delta \downarrow 0$, but not as the contour approaches the principal pole line $\operatorname{Re} s = 1$. No zero-free hypothesis appears.

11.6 The separately scoped unitary infinity type

Let $\nu(\mathfrak{a}) = \nu_0(\mathfrak{a})(N\mathfrak{a})^{-i\tau}$ be fixed and unitary, with angular parameter m and gamma factor γ_ν stated in Section 10. The external primitive functional equation is explicitly

$$Q_\rho^{s/2} \gamma_\nu(s) L(s, \rho) = \epsilon_\rho Q_\rho^{(1-s)/2} \gamma_{\bar\nu}(1 - s) L(1 - s, \bar\rho), \qquad |\epsilon_\rho| = 1.$$

A constant norm-twist phase in another completion convention is absorbed in ϵ_ρ. All finite Kummer twists share this infinity type. Replace T in the kernel by $T_\nu = 1 + |t + \tau| + |m|$. The numerator gamma ratio now has real base $1/2 + |m|/2 > 0$ and vertical parameter $t + \tau$; its conjugate dual has parameter $-(t + \tau)$. Stirling gives the same integrable Gaussian-majorized kernel on each fixed shifted line. Constants can depend on m, τ; T_ν is comparable to T for this one fixed ν. The finite-prime conductor analysis, modulus-one coefficient twist, all-ideal decomposition and all original primitive row values remain unchanged.

A pure norm row has $L(s, \rho) = \zeta_K(s + i\tau)$; exclude it from the pole-free approximate functional equation and use scalar convexity at $t + \tau$. In the Poisson step cancel its actual pole by

$$h_\tau(s) = \frac{s - 1 + i\tau}{s + 1 + i\tau}.$$

This factor is analytic in the relevant right half-plane. On its boundary $|h_\tau| \le 1$, while inside $|h_\tau^{-1}| \le (3/2 + \delta)/(1/2 - \delta)$. Hence the same transfer factor 200 applies to the now ν-dependent central constant. If the opposite norm convention is chosen, replace τ by $-\tau$ in the gamma factor and pole alike. This proves the separate fixed-unitary moment assertion; it does not claim uniformity in a family of infinity types or replace the finite-order principal-residue expression below.

11.7 Keeping the full type-I mask during transfer

For $\sigma = 1/2 + \delta$, define L^T by removing every prime of $T(u, d)$, not just conductor primes. The identity (3) is equivalently

$$F_{u,d}(s) = \frac{L(s, \psi_u)}{L^{T(u,d)}(2s, \psi_{u,2})} \prod_{p \in T(u,d),\ p \nmid \mathfrak{f}_u} (1 - \psi_u(p)(Np)^{-s}).$$

Absolute convergence at $2\sigma > 1$ bounds the reciprocal incomplete product by $\zeta_K(2\sigma)$, without requiring any zero-free theorem in the critical strip. For every fixed $\eta > 0$,

$$\prod_{p \in T} (1 + (Np)^{-\sigma}) \ll_{\sigma,\eta} (N \operatorname{rad} T)^\eta, \qquad N \operatorname{rad} T \le C_S Nu\, Nd.$$

Indeed $1 + q^{-\sigma} \le q^{\eta}$ beyond a fixed threshold, and only finitely many prime ideals lie below it. Therefore

$$|F_{u,d}(\sigma + it)| \ll_{\nu,S,\delta,\eta} (Nu\,Nd)^{\eta} |L(\sigma + it, \psi_u)|. \tag{13}$$

This is a bounded inequality after the exact mask identity, not an identity that removes original zeros. For example, with principal ν and $u = \pi^3$ or π^6 at $N\pi = 7$, the primitive local ratio, restoration and full masked factor at $s = 1$ are respectively

$v_\pi(u)$	primitive ratio	restoration	masked factor
3	48/49	49/48	1
6	8/7	7/8	1

The squared primitive character in the first row is unramified at π although the original primitive character is ramified there. Squaring a zero sentinel would give the wrong denominator.

For finite-order ν, the normalized type-I piece is the retained d sum of the smoothed squarefree series at $Y = D/Nd$, divided by $\sqrt{D}$. Its possible pole is only at $s = 1$, on principal rows. Contour shifting with the full mask gives $I_u = P_u + E_u$, with E_u exactly (4) and

$$P_u = \mathbf{1}_{\psi_u = 1} \sqrt{D}\, \widehat{W}(1) \sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{Nd} \rho_{T(u,d)},$$

$$\rho_T = \frac{\operatorname{res}_{s=1} \zeta_K(s)}{\zeta_K(2)} \prod_{p \in T} (1 + (Np)^{-1})^{-1}.$$

This does not impose positivity on P_u. Since β_U is supported on $Nd \le U^2 \le D$, every inner scale Y is at least one. Four integrations by parts give $|\widehat{W}(\sigma + it)| \ll_\sigma \mathcal{M}_4(W)(1 + |t|)^{-4}$, which justifies the shifted integrals and their horizontal limits with the classical Hecke growth estimates.

Apply Minkowski to (4) in ℓ^2 of the actual u rows, then use (13) and the off-central moment. For sufficiently small internal losses this bounds $\|E\|_2$ by

$$CD^{\delta} H^{1/2 + O(\eta)} \sum_d \frac{|\beta_U(d)|}{(Nd)^{\sigma - \eta}} \int_{\mathbb{R}} |\widehat{W}(\sigma + it)| (1 + |t|)^{(1 + O(\eta))/2}\, dt.$$

All original $\nu(d)\chi_d(u)$ have modulus at most one and remain inside the exact expression before this inequality. The actual truncated factorization gives, without replacing β_U by arbitrary coefficients,

$$\sum_d \frac{|\beta_U(d)|}{(Nd)^{\sigma - \eta}} \le \left(\sum_{Na \le U} (Na)^{-\sigma + \eta} \right)^2 \ll_{\delta,\eta} U^{1 - 2\delta + 2\eta}.$$

Choose $\eta < \delta$ if necessary and use the fixed-field ideal count. The height integral is bounded by $C\mathcal{M}_4(W)$. Squaring and absorbing the small losses in $(HD)^{\varepsilon}$, using $U^2 \le D$, proves (5). Thus the formerly conditional numerator moment input in this residual transfer is discharged, while its positive-power cutoff loss is retained explicitly.

11.8 What remains open and what has been verified

The new remainder envelope improves the old scalar-convexity H power $3/2 - \delta + \varepsilon$ to $1 + \varepsilon$. At fixed U one may choose δ for a requested small-loss exponent; for $U = D^a$ with fixed $a > 0$, the remaining excess tends to $2a$ as $\delta \downarrow 0$. Neither statement controls P_u, the other signed piece, nor their covariance. In the original Vaughan identity the short $2a_U$ term must be restored at smaller

scales where its support can meet the annulus. Uniform $\mathcal{M}_4$ bounds do not justify interchanging a profile supremum with the row moment or assuming $U^2 \le D'$ after a scale D' is reduced.

The independent prose audit is recorded in `planning/analytic-oracle/arithmetic-review/adversarial-afm-review-20261104.md`. The source-hashed finite checker tests valuation/support decompositions, squarefull coordinates, height exponents and rational Poisson envelopes. Its finite success does not prove reciprocity, analytic continuation, the large sieve, the approximate functional equation or the moments. No new Lean declarations accompany these analytic theorems. Existing Lean local Euler identities are a separately limited verification layer. The independent positive arithmetic determinant comparison, original signed low-gcd short-row estimate, all required contour uniformities and GRH remain open; Dirichlet, Hecke, Dedekind and general Artin scopes are not identified with one another.

Pinned primary sources and external inputs. The principal analytic family input is Blomer–Goldmakher–Louvel, arXiv:1112.1650v1, Theorem 1.3, Lemma 2.1, Appendix II and Section 5. Our fixed-ν proof changes column coefficients and redoes the functional-equation reduction; it does not claim the published corollary directly states every fixed twist. For comparison, Gao–Zhao, arXiv:2201.01885v2, Lemma 2.9 and Section 3.2, uses the sextic large-sieve/moment interface. Its GRH-conditional central-density theorem is not invoked. Abelian Kummer identification uses the classical reciprocity theorem, with the project's pinned Milne *Class Field Theory*, version 4.03 (6 August 2020), Theorem 0.8 and Chapter I, Section 1, a reference: Milne's notes. The primitive degree-one Hecke continuation and completed functional equation are explicit classical external inputs; the project's pinned Kudla notes on Tate's thesis, Theorem 4.3 and Corollary 4.4, supply a reference. Their sketch and incomplete displayed epsilon formulas are not used as independently certified constants. Standard Stirling, Mellin inversion, Poisson representation and scalar convexity complete the external analytic dependencies. None is replaced by a finite diagnostic or a GRH-equivalent assumption.

12 Recombining the actual weighted cutoff

Keep the finite-order Eisenstein family and original masks of Section 10. For $U \ge 1$ and a squarefree ideal m, put

$$A_U(m) = \sum_{d \mid m} \beta_U(d) = \sum_{\substack{ab \mid m \\ (a,b)=1,\ Na, Nb \le U}} \mu(a)\mu(b).$$

Every summation variable here is an integral ideal. The last formula uses squarefreeness of m; an unconstrained square of a Möbius polynomial would also include forbidden overlapping supports.

Theorem 5 (Exact recombination and an absolute-half-plane rate)**.** *Write $T_0 = S \cup \operatorname{supp}(u)$ and $J_{u,U}(s) = \sum_{d \text{ squarefree},\ (d,S)=1} \beta_U(d)\nu(d)\chi_d(u)(Nd)^{-s}F_{u,d}(s)$. For $\operatorname{Re} s > 1$,*

$$J_{u,U}(s) = \sum_{m \text{ squarefree},\ (m,T_0)=1} A_U(m)\psi_u(m)(Nm)^{-s},$$
$$|A_U(m)| \le 3^{\omega(m)}, \qquad A_U(m) = \mu(m) \quad (Nm \le U).$$

For every $\kappa > 0$ and $0 < \epsilon < \kappa$, uniformly for $\operatorname{Re} s \ge 1 + \kappa$, all original rows and all $U \ge 1$,

$$\left| J_{u,U}(s) - \frac{1}{L^{T_0}(s, \psi_u)} \right| \le U^{-\kappa+\epsilon}\{\zeta_K(1+\epsilon)^3 + \zeta_K(1+\epsilon)\}.$$

Here L^{T_0} removes every prime of T_0, including conductor-drop primes; the bound is in the absolute-convergence half-plane only.

Proof. On a surviving term $m = dc$, the two squarefree ideals c, d are coprime, and both avoid T_0. The original product $\nu(d)\chi_d(u)\psi_u(c)$ is then $\psi_u(m)$. Grouping by m gives A_U. Each prime of m has three possible positions in $(a, b, m/(ab))$, so the absolute sum is at most $3^{\omega(m)}$. When $Nm \le U$ all divisors fit both cutoffs; the primewise sum is $1 - 1 - 1 = -1$, giving $\mu(m)$.

The same bound justifies absolute convergence and regrouping for $\operatorname{Re} s > 1$, because on squarefree ideals $3^{\omega(m)} = \tau_3(m)$ and $\sum_m \tau_3(m)(Nm)^{-v} = \zeta_K(v)^3$ for $v > 1$, where this latter sum runs over all nonzero integral ideals. Subtract the absolutely convergent Möbius series for $1/L^{T_0}$. Only $Nm > U$ remain. Rankin's inequality with exponent $\eta = \kappa - \epsilon$ bounds their absolute sum by

$$U^{-\eta} \sum_m \{\tau_3(m) + 1\}(Nm)^{-1-\epsilon},$$

which proves the stated constant independently of deleted primes, row multiplicities or character phases. □

For W supported in $[1, 2]$, the same exact regrouping gives the original normalized type-I polynomial

$$I_u = D^{-1/2} \sum_{m \text{ squarefree},\ (m,S)=1} A_U(m)\nu(m)\chi_m(u)W(Nm/D).$$

This formula retains all original zero sentinels. It permits a direct family large sieve before any absolute d sum; its quantitative moment is developed separately below. The displayed absolute-half-plane rate alone is not that critical-strip moment.

13 Euler packaging, coprimality and the analytic boundary

For $p \notin T_0$, set $z_p = \psi_u(p)(Np)^{-s}$ and $t_p = z_p/(1 + z_p)$. On $\operatorname{Re} s > 1$, deleting an additional surviving prime divides the squarefree Euler product by $1 + z_p$. Consequently the exact retained expression is

$$J_{u,U}(s) = F_{u,1}(s)B_{u,U}(s), \qquad B_{u,U}(s) = \sum_{d \text{ squarefree},\ (d,T_0)=1} \beta_U(d) \prod_{p|d} t_p.$$

Original zeros enforce $(d, u) = 1$ even when a sixth power drops from its primitive conductor. This factorization cannot restore a forbidden d row by replacing a zero sentinel with an unramified primitive value. The equality continues wherever both sides are holomorphic.

Without the cutoffs, $\beta_\infty(d) = \mu(d)2^{\omega(d)}$ on squarefree ideals. Absolute convergence justifies

$$B_{u,\infty} = \prod_{p \notin T_0} (1 - 2t_p) = \prod_{p \notin T_0} \frac{1 - z_p}{1 + z_p}, \qquad F_{u,1}B_{u,\infty} = \prod_{p \notin T_0} (1 - z_p) = \frac{1}{L^{T_0}(s, \psi_u)}.$$

Theorem 5 gives a quantitative rate rather than an unsupported exchange of this limit with a critical-strip contour.

A local counterexample to the naive square. In a one-prime projection at a good Eisenstein prime of norm 7, take the principal local phase and $s = 1$. Then $z = 1/7$, $t = 1/8$ and, for $U \ge 7$, the exact polynomial is $B_U = 1 - 2t = 3/4$. The unconstrained square $(1 - t)^2 = 49/64$ differs by $1/64$: it admits the forbidden pair $a = b = p$. After multiplying by $F = 1 + z = 8/7$ it gives $7/8$ instead of the correct $6/7$. This is an actual local Euler calculation, not an infinite-series or mean-square counterexample. It identifies the exact coprimality condition that a proposed mollifier-square simplification must preserve.

The exact intermediate coefficient band. For a squarefree m with $U < Nm \le U^2$, one has $A_U(m) = -\mu(m)$. Indeed the unrestricted disjoint-assignment sum is $\mu(m)$. If $Na > U$, every $b \mid m/a$ automatically has $Nb \le Nm/Na < U$, and $\sum_{b|m/a} \mu(b)$ vanishes unless $a = m$. Hence the excluded a assignments contribute exactly $\mu(m)$; the excluded b assignments contribute another $\mu(m)$. They cannot overlap because $NaNb > U^2 \ge Nm$ contradicts $ab \mid m$. Subtraction proves the claim. Together with the small-norm band, this gives $A_U = \mu$ for $Nm \le U$ and $A_U = -\mu$ for $U < Nm \le U^2$. The original smoothed columns have $Nm \ge D \ge U^2$, so the new band does not justify replacing their coefficients by $-\mu(m)$: generally they lie beyond it.

Theorem 6 (A stronger cutoff convergence would require zero-freeness)**.** *Fix a nonprincipal primitive ψ_u. If these actual $J_{u,U}$ converge locally uniformly on $\operatorname{Re} s > 1/2$ as $U \longrightarrow \infty$, then $L(s, \psi_u)$ has no zero there.*

Proof. Classical abelian Hecke continuation makes the numerator of $F_{u,1} = L^{T_0}(s, \psi_u)/L^{T_0}(2s, \psi_{u,2})$ holomorphic. Its denominator is nonzero on $\operatorname{Re} s > 1/2$ by the absolute Euler product at $2s$. Each $B_{u,U}$ is a finite sum with nonzero denominators $1 + z_p$ on $\operatorname{Re} s > 0$. Thus every $J_{u,U}$ is holomorphic on the asserted half-plane. A locally uniform limit G is holomorphic there. On $\operatorname{Re} s > 1$, the already proved convergence gives $L^{T_0}(s, \psi_u)G(s) = 1$. The identity theorem extends this equality to the connected half-plane, precluding a zero of L^{T_0}. The finitely deleted factors $1 - \psi_u(p)(Np)^{-s}$ do not vanish when $\operatorname{Re} s > 0$, so the same conclusion holds for the primitive L-function. □

This is an implication, not a proof that the stronger convergence holds, and it does not assert a converse. In particular, a family-averaged energy bound is not silently promoted to locally uniform convergence for each individual character. Principal rows have a pole at 1 and were excluded from this proposition. General Artin holomorphy is not an input.

Reproducible exact diagnostics. The new source `examples/arithmetic_cutoff/exact_recombination.py` uses actual split/inert prime ideals and raw sextic symbols in $\mathbb{Z}[\omega]$, with exact rational arithmetic. It checks the direct masked sum, coefficient regrouping, Euler packaging, both finite full-cutoff limits and original zeros, including third/sixth-power rows. Its overlapping- support negative controls are not tests of an infinite large sieve or of zero location. The analytic convergence argument above uses ordinary ideal Euler products and the explicitly stated Hecke continuation theorem; no new Lean declaration or complete GRH proof is claimed.

14 A cutoff-uniform envelope for the original masked remainder

Fix $K = \mathbb{Q}(\omega)$, $\mathcal{O} = \mathbb{Z}[\omega]$, a finite-order Hecke character ν, and a finite set S containing the primes above 6 and the conductor of ν. All unspecified ideal sums below are over nonzero integral ideals. Let $\chi_m(u) = (u/m)_6$ be the original symbol, zero when $(m, u) > 1$, and put

$$\beta_U(d) = \sum_{\substack{ab=d \\ Na, Nb \le U}} \mu(a)\mu(b), \qquad A_U(m) = \sum_{d|m} \beta_U(d).$$

The columns d, c, m are squarefree and prime to S where indicated. Retain the primitive characters $\psi_u = \operatorname{prim}(\nu\kappa_u)$, $\psi_{u,2} = \operatorname{prim}(\psi_u^2)$, and the full deleted set $T(u, d) = S \cup \operatorname{supp}(u) \cup \operatorname{supp}(d)$. For $H, D, U \ge 1$, $U^2 \le D$, and $W \in C_c^\infty((0, \infty))$ supported in $[1, 2]$, the original normalized type-I piece is

$$I_u = D^{-1/2} \sum_{\substack{m \text{ squarefree} \\ (m,S)=1}} A_U(m)\nu(m)\chi_m(u)W(Nm/D). \tag{14}$$

Its exact principal residue is

$$P_u = \mathbf{1}_{\psi_u=1}\sqrt{D}\,\widehat{W}(1) \sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} \frac{\beta_U(d)\nu(d)\chi_d(u)}{Nd}\rho_{T(u,d)},$$

$$\rho_T = \frac{\operatorname{res}_{s=1}\zeta_K(s)}{\zeta_K(2)} \prod_{p\in T}(1+(Np)^{-1})^{-1}. \tag{15}$$

Define $E_u = I_u - P_u$; this is the same residue-subtracted remainder as the fully masked Mellin contour, with no cutoff limit or profile supremum.

Theorem 7 (Uniformity in the actual truncated cutoff)**.** *For every $\varepsilon > 0$,*

$$\sum_{0<Nu\le H} |E_u|^2 \ll_{\nu,S,\varepsilon} \|W\|_\infty^2 (HD)^\varepsilon \{H + DH^{1/3} + (HD)^{2/3}\}. \tag{16}$$

The sum is over every actual nonzero Eisenstein element, including units, S-supported powers and sixth-power rows. The constant is independent of U under $U^2 \le D$. This is an application of the classical sextic large sieve with exact original coefficients, not a zero-location theorem.

Proof. On an absolutely convergent line, expand the restored squarefree series $F_{u,d}$. Its condition $(c, T(u,d)) = 1$ includes $(c,d) = 1$, and the original symbols multiply. Grouping $m = cd$ gives (14) by Mellin inversion. Each prime of m may belong to the first truncated Möbius divisor, to the second, or to neither. Consequently $|A_U(m)| \le 3^{\omega(m)} \ll_\eta (Nm)^\eta$, uniformly in U. The length-D coefficient sequence in (14) therefore has squared norm $\ll_\eta \|W\|_\infty^2 D^\eta$.

The all-ideal row norm in Section 3 of Blomer–Goldmakher–Louvel, arXiv:1112.1650v1 gives

$$B_2^1(X,D) \ll_\eta (XD)^\eta \{X + DX^{1/3} + (XD)^{2/3}\}.$$

For completeness, split $a = a_1 a_2^2 \cdots a_5^5 a_6^6$ dyadically; $a_1, \dots, a_5$ are squarefree and pairwise coprime. If their dyadic sizes are A_j, put $R = \prod_{j=1}^6 A_j$ and $M = \max(A_1, A_2)$. Pivot on the first or second variable, using the squarefree powered sieve proved simultaneously in that paper's Section 3.2. The other variables are bounded column multipliers. The block energy is at most

$$(XD)^\eta \{R + DR/M + D^{2/3}R/M^{1/3}\} \sum_m |\lambda_m|^2.$$

The inequalities $R \le X$, $R^3 \le XM^3$ and $R^3 \le X^2 M$ follow from $\prod_j A_j^j \le X$ and the pivot choice. Summing the logarithmically many blocks proves the displayed bound.

Finite S-unit and ray-class partitions convert the original literal symbol into the canonical column value $\vartheta_m(a)$ with fixed modulus-one coefficient multipliers. On noncoprime pairs both values are zero: squarefree m has outside-S conductor exactly m. This orientation preserves sixth-power zeros. At most six elements generate each outside-S ideal times an S-supported ideal; the S-supported sums converge at powers 1, 1/3 and 2/3. Thus (16) holds first with I_u in place of E_u.

A principal ψ_u requires every outside-S valuation of u to be divisible by 6. There are $O_{\nu,S}(H^{1/6})$ such literal rows. Equation (15), $|\rho_T| \ll_K 1$, and the actual coefficient sum

$$\sum_d \frac{|\beta_U(d)|}{Nd} \le \left(\sum_{Na\le U} \frac{1}{Na}\right)^2 \ll \log^2(2U)$$

give $\sum |P_u|^2 \ll DH^{1/6}\|W\|_\infty^2 \log^4(2U)$. This is absorbed by the right side of (16), using $U^2 \le D$ and a smaller internal exponent. Finally $|E_u|^2 \le 2|I_u|^2 + 2|P_u|^2$. The extended proof specifies the canonical-family transfer and the powered-sieve input precisely. □

Combining this with the previously proved fixed-twist AFM transfer yields

$$\sum_{0<Nu\le H} |E_u|^2 \ll_{\nu,S,\delta,\varepsilon} \mathcal{M}_4(W)^2 (HD)^{\varepsilon} \min\{D^{2\delta}U^{2-4\delta}H,\ H + DH^{1/3} + (HD)^{2/3}\}. \tag{17}$$

At $H = D^{3/4}$ and $U = D^a$, $0 \le a \le 1/2$, the excess over H is at most $\min\{2\delta + (2-4\delta)a, 1/2\}$, up to arbitrarily small losses. The new envelope improves the old one for $a > 1/4$ as $\delta \downarrow 0$, but still leaves an exponent loss of $1/2$. It does not settle the original signed short-row bound, type-II covariance, positive arithmetic determinant comparison or GRH. The principal-residue bound used here is sufficient for this envelope, not the stronger target needed to reassemble the full signed argument.

15 All-ideal reciprocity and the surviving block obstruction

The proof uses no new zero-free input. Its external analytic ingredient is the pinned BGL large sieve and the powered estimates established in its proof; elementary ideal counting and the already restored Mellin identity supply the remaining interfaces. All constants may depend on fixed K, S, ν and declared small exponents, never on U.

15.1 Exact recombination retains the primitive corrections

On $\operatorname{Re} s > 1$, start from the full defining series, with its original zero masks:

$$F_{u,d}(s) = \sum_{\substack{c \text{ squarefree} \\ (c,T(u,d))=1}} \nu(c)\chi_c(u)(Nc)^{-s}.$$

Its primitive ratio involves $L(s,\psi_u)$, the independently primitive $L(2s,\psi_{u,2})$ and every deleted Euler factor. Expanding this *restored* series gives the exact equality

$$\begin{aligned} &\sum_{\substack{d \text{ squarefree} \\ (d,S)=1}} \beta_U(d)\nu(d)\chi_d(u)(Nd)^{-s}F_{u,d}(s) \\ &\quad = \sum_{\substack{m \text{ squarefree} \\ (m,S)=1}} A_U(m)\nu(m)\chi_m(u)(Nm)^{-s}. \end{aligned} \tag{18}$$

Indeed $(c,d) = 1$ makes $m = cd$ squarefree, with $d \mid m$. If m meets u, both original symbols retain their zero. Only on the absolutely convergent line is this infinite series used. Mellin inversion there gives the finite smoothed polynomial (14). Nothing evaluates (18) as a Dirichlet series on $\operatorname{Re} s = 1/2 + \delta$ or interchanges a cutoff limit with continuation across unknown zeros.

For squarefree m, a factorization $ab = d \mid m$ assigns each prime to a, to b, or to neither. The actual inequalities $Na, Nb \le U$ select a subset of these assignments. Thus

$$|A_U(m)| \le 3^{\omega(m)} \ll_\eta (Nm)^\eta.$$

The last divisor bound follows by choosing a fixed threshold so $3 \le (Np)^\eta$ beyond it, then absorbing the finitely many smaller prime ideals in a constant. When $Nm \le U$ all assignments occur, so $A_U(m) = \prod_{p\mid m}(1-1-1) = \mu(m)$; no such simplification is made for the length-D columns in general. With $\lambda_m = D^{-1/2}A_U(m)\nu(m)W(Nm/D)$, ideal counting and a smaller internal η give

$$\sum_m |\lambda_m|^2 \ll_\eta \|W\|_\infty^2 D^\eta, \qquad \operatorname{supp}\lambda \subseteq \{D \le Nm \le 2D\}. \tag{19}$$

15.2 The powered squarefree input and all-ideal pivot

Let ϑ_a be BGL's canonical primitive sixth-order family. Their Section 3 defines B_1^j using squarefree row/column ideals and kernel $\vartheta_a(m)^j$, while B_2^j uses *all* ideal rows and kernel $\vartheta_m(a)^j$. Theorem 1.3 states the squarefree bound for $j = 1$; its Section 3.2 proves the estimates simultaneously for $1 \le j \le 5$, because the recursion in Lemma 3.4 uses $j\ell$. In particular the two inputs needed here are

$$B_1^j(M, D) \ll_\eta (MD)^\eta\{M + D + (MD)^{2/3}\}, \qquad j = 1, 2. \tag{20}$$

To see the final step in the published recursion explicitly, its $E_{4/3}$ estimate is $M^{4/3} + D + (MD)^{2/3}$. If $M \le D$, its first term is at most $(MD)^{2/3}$; if $M > D$, use reciprocity/duality (3.2) and the estimate with M, D reversed. The finite set of powers is treated together; no inference from a fixed conductor count to an arbitrary primitive-character sieve occurs.

Write each ideal row uniquely as

$$a = a_1a_2^2a_3^3a_4^4a_5^5a_6^6,$$

where $a_1, \dots, a_5$ are pairwise-coprime squarefree ideals and a_6 is arbitrary. For valuation $6q + r$, $0 \le r \le 5$, place the prime in a_r if $r > 0$ and its power q in a_6. Dyadically let $A_j \le Na_j < 2A_j$, with $A_j \ge 1$, and put $R = \prod_{j=1}^6 A_j$. A nonempty block satisfies $\prod_j A_j^j \le X$ when $Na \le X$. Choose $\ell = 1$ if $A_1 \ge A_2$, and $\ell = 2$ otherwise; put $M = A_\ell$. Keep every other ideal fixed. Evaluation of the single *column* character gives

$$\vartheta_m(a) = \vartheta_m(a_\ell)^\ell\vartheta_m(a/a_\ell^\ell).$$

The second factor is a bounded column multiplier, including its original zero if m meets any fixed factor. It may depend on all the fixed ideals but not on the pivot row. Pairwise-coprimality and the exact row norm restriction merely select a subset of the positive pivot-row sum. Apply (20) to the first factor. There are $O_K(R/M)$ choices of the other ideals by ideal counting. The block energy is bounded by

$$(XD)^\eta\{R + DR/M + D^{2/3}R/M^{1/3}\}\sum_m |\lambda_m|^2. \tag{21}$$

All three monomial bounds used to sum these blocks are elementary. First $R \le \prod_j A_j^j \le X$. If $M = A_1$, then

$$(R/M)^3 = A_2^3 \cdots A_6^3 \le A_1A_2^2A_3^3A_4^4A_5^5A_6^6 \le X,$$

since $A_2 \le A_1$ and $j \ge 3$ for the remaining factors. If $M = A_2$, use $A_1 \le A_2$ in the same comparison. Finally,

$$\frac{R^3}{M(\prod_j A_j^j)^2} = \frac{A_1}{MA_2A_3^3A_4^5A_5^7A_6^9} \le 1.$$

Thus $R/M \le X^{1/3}$ and $R/M^{1/3} \le X^{2/3}$. Insert these in (21) and sum the logarithmically many blocks. This proves the all-ideal estimate used in Theorem 7. It is precisely the mechanism of BGL Lemma 3.4 and (3.6)–(3.9), with the final powered bound inserted. The simpler fixed-squarefull-row argument would leave the weaker $D\sqrt{X}$ term.

15.3 Literal rows, reciprocal orientation and principal count

Let $a = (u)_S$ denote the outside-S ideal. The finite S-unit comparison gives $\kappa_u = \eta_u\vartheta_a$ as primitive characters, with η_u in a finite fixed set. For $(a, m) = 1$, BGL reciprocity therefore gives

$$\chi_m(u) = \eta_u(m)C([a], [m])\vartheta_m(a). \tag{22}$$

The notation is deliberate: the left side is the project's original $(u/m)_6$, not BGL's canonical character indexed by m. When $(a,m)>1$, the left side is zero. The right side is also zero, because squarefree m has outside-S primitive conductor exactly m. Equation (22) hence extends to every pair. Using $\vartheta_a(m)$ on noncoprime pairs instead would be false: primitive ϑ_{b^6} has restored primes of b, whereas the original symbol must still vanish there.

Partition rows by η_u and $[a]$, both finite. On each part the factor $\eta_u(m)C([a],[m])$ is a fixed modulus-one column multiplier. A positive row sum over that part is bounded by the full B_2^1 row sum. For each S-supported ideal b and outside-S ideal a, at most six actual elements generate ab. Thus the required literal energy is bounded by the all-ideal norm summed over b, with $X=H/Nb$. The geometric products

$$\sum_{b\ S\text{-supported}} (Nb)^{-q} = \prod_{p\in S}(1-(Np)^{-q})^{-1} < \infty, \qquad q = 1,\ 1/3,\ 2/3,$$

transfer the three terms without losing any unit or S-power row. Together with (19), this proves the claimed I energy.

If ψ_u is principal, ν is unramified outside S, and tame Kummer inertia at $v_p(u)$ has order $6/\gcd(6,v_p(u))$. It must be trivial, so each such valuation is divisible by 6. Writing $(u)=ba_6^6$ with b supported in S, ideal counting gives

$$\#\{0<Nu\le H:\psi_u=1\} \ll_K H^{1/6}\prod_{p\in S}(1-(Np)^{-1/6})^{-1}.$$

This is only a necessary condition, sufficient for an upper count. The exact residue formula (15) and $|\widehat{W}(1)|\le\|W\|_\infty$ then give the principal energy bound used above. No sign or positivity of that residue is assumed.

15.4 What the envelope still fails to prove

The $DH^{1/3}$ term is still larger than $H(HD)^{\varepsilon}$ at $H=D^{3/4}$. The monomial comparison itself is saturated by $A_1=A_2=X^{1/3}$ and $A_3=\cdots=A_6=1$. This proves sharpness of that *particular comparison*, not an energy lower bound for actual arithmetic coefficients. In particular, no conclusion that the signed A_U cannot cancel is justified. The next load-bearing input must exploit those actual truncated Möbius coefficients inside the mixed moment, or establish the required signed type-II minus type-I covariance. The smaller-scale short term, principal contribution, full arithmetic determinant identification and GRH remain separate obligations.

Exact finite checks use the split Eisenstein primes generated by $3+\omega$, $4+\omega$, $5+2\omega$, $6+\omega$, of norms $7,13,19,31$. Their actual residue symbols are computed in $\mathbb{Q}(\zeta_6)$, retaining sixth-power zeros. Coefficient recombination and dyadic inequalities pass; they check finite algebra and bookkeeping, not the external analytic sieve or the contour theorem. No new Lean declarations accompany this argument.

16 High cutoffs, actual bilinear factors and a signed covariance test

Retain the original Eisenstein family, fixed finite-order ν, fixed S, common $W\in C_c^\infty((0,\infty))$ supported in $[1,2]$, and every raw shared-prime zero. Put

$$C_U(m)=\sum_{\substack{abc=m\\ Na>U,\ Nb>U}}\mu(a)\mu(b), \qquad r_U(m)=\#\{(a,b):ab=m,\ Na,Nb>U\}.$$

All factors are ideals, squarefree and pairwise coprime; on the active annulus $A_U = C_U - \mu$. A coefficient identity is different from a row-energy bound.

Proposition 8 (The high-cutoff window)**.** *If $D > 1$ and $2D/7 \le U^2 \le D$, then for every surviving squarefree column $D \le Nm \le 2D$,*

$$C_U(m) = \mu(m) r_U(m), \qquad A_U(m) = \mu(m)(r_U(m) - 1), \qquad |A_U(m)| \ge 1.$$

For each fixed nonzero W, the normalized squared coefficient norm satisfies

$$D^{-1} \sum_{\substack{m \text{ sf} \\ (m,S)=1}} |A_U(m) W(Nm/D)|^2 \gg_{W,S} 1 \tag{23}$$

for large D, uniformly in this window. This is not a lower bound for row energy.

Proof. The least norm of a nonunit ideal outside S is at least 7. A long-long term with $c \ne 1$ would have $Nm > 7U^2 \ge 2D$. Thus $ab = m$, including both endpoints. The fixed-point-free swap $(a, b) \mapsto (b, a)$ makes r_U even, so A_U is an odd nonzero integer. The last claim follows by counting squarefree ideals on a fixed interval where $|W|$ is bounded below; the constant and error are proved in the extended proof. □

The exact remaining polynomial is genuinely bilinear, with $(a, b) = 1$, $U < Na, Nb < 2D/U$, $D \le Na\, Nb \le 2D$, and $x_{ab}(u) = x_a(u) x_b(u)$, including zeros. Both factors see the same row $u = \varepsilon\alpha\beta^2$. Cauchy followed by the accepted powered BGL bound gives only

$$Q_{B_X}(C_U, C_U) \ll_{\nu,S,\epsilon} (DX)^{\epsilon} \|W\|_\infty^2 \{\sqrt{D}\, X^2 + DX + D^{5/6} X^{5/3}\}. \tag{24}$$

At $X \asymp D^{1/4}$ the exponents are $1, 5/4, 5/4$; the earlier regrouped bound has $1/2, 5/4, 13/12$ and is stronger. No BT_σ saving is obtained. Multiplying two marginal operator norms does not supply a coupled fourth moment.

A finite test in the original family. There is no automatic nonnegative signed covariance, even when $C_U(m)\mu(m) \ge 0$. Take $\nu = 1$, S the primes above $2, 3$, $D = 400$, $U = 17$, and all 72 literal rows of B_7. Choose the common nonnegative smooth profile constructed in the extended proof, with values 10 at 439/400, 1 at 475/400, and zero at every other integer norm divided by 400. There are exactly four active squarefree ideal columns, of norms $439, 439, 475, 475$. In that order,

$$A = (1, 1, 1, 1), \quad C = (0, 0, 2, 2), \quad \mu = (-1, -1, 1, 1), \qquad \Re G_{B_7} = \begin{pmatrix} 72 & 0 & 0 & 18 \\ 0 & 72 & 18 & 0 \\ 0 & 18 & 72 & 0 \\ 18 & 0 & 0 & 72 \end{pmatrix}.$$

Exact raw-symbol enumeration, independently reproducible from the ideal and residue-field definitions, gives

$$Q(C, \mu) = -\frac{27}{25}, \qquad Q(C, C) = \frac{36}{25}, \quad Q(\mu, \mu) = \frac{864}{25}, \quad Q(A, A) = \frac{954}{25}. \tag{25}$$

The real Gram matrix is positive definite, with eigenvalues $54, 90$; its positivity does not force a cross covariance to be positive. This fixed finite example refutes that shortcut, not the asymptotic BT_σ target or the original signed mean-square estimate.

The two-cutoff contour remains coupled. For $T_0 = S \cup \operatorname{supp}(u)$, $\psi = \psi_u$ and independently primitive $\psi_2 = \operatorname{prim}(\psi^2)$, put $q_p = \psi(p)(Np)^{-s}$, $\alpha_p = (Np)^{-z}$, $\beta_p = (Np)^{-w}$ outside T_0. The actual three-placement Euler factor is $1 + q_p - q_p\alpha_p - q_p\beta_p$. Its exact factorization is

$$\mathcal{F}(s,z,w) = \frac{L^{T_0}(s,\psi)}{L^{T_0}(s+z,\psi)L^{T_0}(s+w,\psi)} \prod_{p \notin T_0} \left(1 - \frac{q_p^2(1-\alpha_p)(1-\beta_p)}{(1-q_p\alpha_p)(1-q_p\beta_p)}\right). \tag{26}$$

Exact double Perron inversion is valid on positive auxiliary lines $c, d > 1 - \Re s$ for $1/2 < \Re s < 1$, retaining the original principal pole exclusion. These safe lines reproduce the previous $U^{2-4\delta}$ energy loss at $\Re s = 1/2 + \delta$, with the same common-profile derivative and fixed-margin dependencies. The correction product alone does not justify moving those lines through reciprocal-L poles. In fact $\mathcal{F}(s,z,0) = 1/L^{T_0}(s+z,\psi)$ as a meromorphic identity. No zero-free region, reciprocal moment, analytic saving, new Lean formalization or GRH proof is supplied by this construction.

17 Proof details for the high-window and two-cutoff tests

17.1 The coefficient lower bound and its precise limitation

The element lattice $\mathbb{Z}[\omega] \subset \mathbb{C}$ has covolume $\sqrt{3}/2$. Counting disks with boundary error $O(\sqrt{x})$ and dividing by the six units, using that this ring is a PID, gives

$$N_K(x) = \kappa x + O(\sqrt{x}), \qquad \kappa = \frac{\pi}{3\sqrt{3}}.$$

Finite inclusion-exclusion at S gives $N_{K,S}(y) = \kappa y \prod_{p\in S}(1-(Np)^{-1}) + O_S(\sqrt{y})$. Insert $1_{\mathrm{sf}}(m) = \sum_{d^2\mid m}\mu(d)$. Absolute convergence at 2, the tail $\sum_{Nd>\sqrt{x}}(Nd)^{-2} = O(x^{-1/2})$, and $\sum_{Nd\le\sqrt{x}}(Nd)^{-1} = O(\log(2x))$ yield

$$N_{\mathrm{sf},S}(x) = c_S x + O_S(\sqrt{x}\log(2x)), \qquad c_S = \frac{\kappa}{\zeta_K(2)} \prod_{p\in S}(1+(Np)^{-1})^{-1} > 0.$$

These tail estimates follow by partial summation from the displayed lattice count. For a fixed nonzero W, choose $[a,b] \subset (1,2)$ with $|W| \ge \eta > 0$. Proposition 8 gives, for large D,

$$D^{-1}\sum |A_U(m)W(Nm/D)|^2 \ge \tfrac{1}{2}\eta^2 c_S(b-a).$$

There is no uniform lower constant over all profiles with one specified supremum norm. Nor can a coefficient lower bound be combined with an operator upper bound to produce a row-energy lower bound: no lower singular value has been established on the actual vector.

17.2 What legal factor Cauchy actually proves

For dyadic $Na \asymp A, Nb \asymp B$, retain the exact $(a,b) = 1$ mask, hyperbolic support and $W(Na\,Nb/D)$ in

$$T_a(u) = \sum_b \mu(b)\nu(b)W(Na\,Nb/D)x_b(u).$$

Cauchy gives

$$|\sum_a \mu(a)\nu(a)x_a(u)T_a(u)|^2 \ll A\sum_a |T_a(u)|^2.$$

For fixed a, fixed second row factor and fixed unit/ray classes, the b coefficients are independent of the pivot row and have squared norm $\ll B\|W\|_\infty^2$. The prior classical power-one/two BGL input

has $\Delta(X,B) = X + B + (XB)^{2/3}$. Summing the $O(A)$ fixed columns and $O(X)$ other row factors gives

$$\mathcal{E}_{A,B} \ll_{\nu,S,\epsilon} (DX)^{\epsilon}\|W\|_{\infty}^{2}\frac{A^2B}{D}X\Delta(X,B) \ll (DX)^{\epsilon}\|W\|_{\infty}^{2}\{AX^2 + DX + D^{2/3}A^{1/3}X^{5/3}\}.$$

In the high window $A, B \asymp \sqrt{D}$, and only boundedly many dyadic pairs occur. This proves (24); exchanging factors or row pivots leaves its DX term. A product of marginal second-moment bounds supplies no estimate for the two transforms on the same (r,s). Any useful saving must control that coupled fourth moment or the signed actual-coefficient covariance, with all masks retained.

17.3 A smooth profile and the exact covariance computation

Let

$$\phi(t) = \begin{cases} \exp(1 - 1/(1-t^2)), & |t| < 1, \\ 0, & |t| \geq 1, \end{cases} \qquad W_x(y) = x\phi(4(400y - 439)) + \phi(4(400y - 475)).$$

For $x \geq 0$ this is nonnegative and smooth; its disjoint supports lie strictly inside $(1,2)$. For every integer n, $W_x(n/400)$ equals x at $n = 439$, 1 at $n = 475$ and zero otherwise. Thus the computation uses exact values of a genuine common C_c^{∞} profile, rather than claiming a finite rational weight is smooth. At $x = 10$, $\|W_x\|_{\infty} = 10$.

The primary ideal generators, in the order used above, are

$$(-23,-18), \quad (-5,18), \quad (10,-15), \quad (25,15),$$

where (a,b) denotes $a + b\omega$. The first two have norm 439. The last two factor as $(-2,3)(-5,0)$ and $(-5,-3)(-5,0)$, with norms $19 \cdot 25$. Since $19, 25 > 17$, the long-factor count is 2 there and zero on the prime columns. All squarefree ideals at both selected norms are included. The complete B_7 uses the four good primes of norms $7, 7, 13, 13$ for distinct ordered row factors and all six units. It has 72 distinct rows, maximum norm 2197; each is nonprincipal by its outside-S Kummer valuation 1 or 2 modulo 6, whose nontrivial tame character has conductor exponent 1.

For the raw Gram $G_{ij} = \sum_{u \in B_7} x_{m_i}(u)\overline{x_{m_j}(u)}$, exact finite-field exponentiation and lattice enumeration give the displayed real matrix. The weighted real vectors are $\ell = (0,0,2,2)$, $v = (-x,-x,1,1)$, and $a = (x,x,1,1)$. Therefore

$$Q(C_U,\mu) = D^{-1}\ell^t(\Re G)v = \frac{36 - 9x}{50}.$$

Its diagonal contribution is $36/50 > 0$ and its off-diagonal contribution is $-9x/50$. For $x = 10$ this is $-27/25$ and gives (25). The identity $(\Re G - 72I)^2 = 18^2 I$ proves the real eigenvalues $54, 90$; these are not being asserted as the eigenvalues of the full complex Gram. The exact source is `examples/arithmetic_cutoff/signed_covariance.py`. A second test retains raw zeros on the 246 rows $0 < Nu \leq 89$ whose outside-S part is nonunit. Its covariance at $x = 10$ is $-69/50$. This subset is guaranteed nonprincipal; it is not called the entire nonprincipal family. Here $89 = \lfloor 400^{3/4} \rfloor$. These are finite counterexamples to automatic covariance positivity, not asymptotic counterexamples to BT_{σ} or GRH.

17.4 Double Perron without a divergent critical-strip series

For $T_0 = S \cup \operatorname{supp}(u)$ retain $F_0(s) = L^{T_0}(s,\psi)/L^{T_0}(2s,\psi_2)$, with the independently primitive square character. Set $t_p = q_p/(1+q_p)$ outside T_0. The finite actual cutoff is

$$J_U(s,u) = F_0(s)B_U(s), \qquad B_U(s) = \sum_{\substack{a,b \text{ sf, } (a,b)=1 \\ (ab,T_0)=1,\ Na,Nb\le U}} \mu(a)\mu(b)\prod_{p\mid ab} t_p.$$

For $\sigma = \Re s > 1/2$, the series

$$\mathcal{G}(s,z,w) = \sum_{\substack{a,b \text{ sf, } (a,b)=1 \\ (ab,T_0)=1}} \mu(a)\mu(b)\prod_{p\mid ab} t_p\,(Na)^{-z}(Nb)^{-w} = \prod_{p\notin T_0}\left(1 - t_p(Np)^{-z} - t_p(Np)^{-w}\right)$$

converges absolutely on $\sigma + \Re z > 1$, $\sigma + \Re w > 1$, uniformly in all vertical parts at fixed positive margins. This follows from $|t_p| \le (Np)^{-\sigma}/(1-(Np)^{-\sigma})$. There $\mathcal{F} = F_0\mathcal{G}$, without evaluating an infinite c series in the critical strip. Initial absolute convergence also proves (26) by the local identity

$$(1-q)(1+q-q\alpha-q\beta) = (1-q\alpha)(1-q\beta) - q^2(1-\alpha)(1-\beta).$$

The correction product is holomorphic on

$$\sigma + \Re z > 0, \quad \sigma + \Re w > 0, \quad 2\sigma + \min(\Re z, 0) + \min(\Re w, 0) > 1.$$

On margins $\sigma + \Re z, \sigma + \Re w \ge \eta > 0$ and last exponent $\ge 1+\epsilon$, its local deviation from 1 is at most $4(1-2^{-\eta})^{-2}(Np)^{-1-\epsilon}$. This proves normal convergence and a phase/deletion-uniform vertical upper bound. Nonvanishing of the correction is not asserted.

For $1/2 < \sigma < 1$, positive lines $c, d > 1-\sigma$, and $V = \lfloor U \rfloor + 1/2$, define

$$\mathcal{I}_{T_1,T_2} = \frac{1}{(2\pi i)^2}\int_{c-iT_1}^{c+iT_1}\int_{d-iT_2}^{d+iT_2} \mathcal{F}(s,z,w)\frac{V^{z+w}}{zw}\,dw\,dz.$$

The half-integer V includes the actual equality cutoff, since $Na \le U$ is exactly $Na < V$. Put $\Delta_V = \inf_{n\in\mathbb{Z}_{\ge 1}} |\log(V/n)| \ge 1/(2V+1)$, and let $\mathcal{A} = \sum_{a,b} |\mu(a)\mu(b)\prod_{p\mid ab} t_p|(Na)^{-c}(Nb)^{-d}$ with the same disjoint support. For $c, T > 0$ and $v > 0$, $v \ne 1$, define

$$K_{c,T}(v) = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} \frac{v^z}{z}\,dz.$$

Finite-height termwise integration of the absolutely convergent $\mathcal{G}$ and the one-variable tail estimate $|K_{c,T}(v) - 1_{v>1}| \le 2v^c/(\pi T|\log v|)$ give

$$|\mathcal{I}_{T_1,T_2} - J_U| \le |F_0|\mathcal{A}V^{c+d}\left\{\frac{2}{\pi T_1\Delta_V} + \frac{2}{\pi T_2\Delta_V} + \frac{4}{\pi^2 T_1 T_2 \Delta_V^2}\right\}. \tag{27}$$

Thus $\mathcal{I}_{T_1,T_2} \to J_U$ as both heights tend independently to infinity, away from the original principal pole. This is an oscillatory limit, not an absolutely integrable infinite vertical integral.

Taking $T_1 = T_2 = V^2$ gives the safe upper bound $|J_U| \ll (Nu)^\eta |L(s,\psi)| V^{c+d}\log^2(2V)$, with constants retaining fixed margins, ν, S, and σ. Choosing c, d close to $1-\sigma$ reproduces $U^{2-2\sigma+\eta}$ in amplitude and $U^{2-4\delta+\eta}$ in energy on $\sigma = 1/2+\delta$. Insertion into the original common-profile Mellin argument retains $M_4(W)^2$ and all previously required height/conductor uniformities. No derivative-free improvement follows from this Perron estimate.

The correction converges even on some negative auxiliary lines, but moving there can cross $z = \rho - s$ or $w = \rho - s$, from zeros of the primitive L-function. The exact meromorphic slice $\mathcal{F}(s, z, 0) = 1/L^{T_0}(s + z, \psi)$ forbids assuming universal cancellation of these poles. At a numerator zero the slice must be understood meromorphically, not by an unjustified pointwise substitution. Sequential shifts have single boundary integrals and crossed-zero residues as well as the kernel double residue; away from s zeros or poles that double residue is $1/L^{T_0}(s, \psi)$. A principal L pole supplies a reciprocal zero, not a reciprocal pole. Uniform control of actual reciprocal integrals/residues or the signed covariance remains a new input. Numerator AFM alone supplies neither.

18 A quantitative covariance budget without a new exponent

Retain the original Eisenstein sextic symbols, fixed finite-order ν, fixed S, one possibly complex common $W \in C_c^\infty$ supported in $[1, 2]$, all squarefree columns prime to S, and every raw shared-prime zero. In the accepted window $2D/7 \le U^2 \le D$, put $r_m = r_U(m)$, $\lambda_m = \mu(m)\nu(m)W(Nm/D)$ and $v_m = \overline{\lambda_m}$. Thus $C_U = \mu r$ and $A_U = \mu(r - 1)$ on the active annulus. For a finite literal row block B, let

$$G_{mn} = \sum_{u \in B} x_m(u)\overline{x_n(u)}, \qquad R = \operatorname{diag}(r_m), \qquad M_j = \sum_m r_m^j |W(Nm/D)|^2 \quad (0 \le j \le 2).$$

The primitive $\psi_u, \psi_{u,2}$ and full deleted sets retain their previous meanings; no conductor-drop value replaces a raw zero.

Proposition 9 (An actual-vector lower budget)**.** *If $0 \le G \le \Lambda_B I$, then*

$$\Re Q_B(C_U, \mu) \ge -\frac{\Lambda_B}{2D}\{\sqrt{M_0 M_2} - M_1\}. \tag{28}$$

In this high window, uniformly in U,

$$\sum_{D \le Nm \le 2D} r_m \ll_K D, \qquad \sum_{D \le Nm \le 2D} r_m^2 \ll_K D \log(2D). \tag{29}$$

Consequently

$$\Re Q_B(C_U, \mu) \ge -C_K \|W\|_\infty^2 \Lambda_B \sqrt{\log(2D)}.$$

The vector inequality is sharp among positive contractions with the specified v, Rv; no such sharpness is asserted for the arithmetic Gram. The moment bound counts actual balanced ideal factorizations, rather than assuming independence of the two row factors.

On B_X, inserting the previously accepted powered-BGL norm gives only

$$\Re Q_{B_X}(C_U, \mu) \ge -C_{\nu,S,\epsilon} \|W\|_\infty^2 (DX)^\epsilon \{X^2 + DX + D^{2/3}X^{5/3}\}.$$

At $X \asymp D^{1/4}$ the exponents remain $1/2, 5/4, 13/12$. At $H = D^{3/4}$ this budget still has the old $D^{1/2}$ excess over H; it supplies no BT_σ saving. A negative-part budget is weaker than $2\Re Q(C_U, \mu) \ge Q(C_U, C_U) + Q(\mu, \mu) - R_\sigma$.

For $S_R = \sqrt{R}$, the exact identity

$$\tfrac{1}{2}(RG + GR) = S_R G S_R + \tfrac{1}{2}[S_R, [S_R, G]]$$

localizes the commutator to unequal factor-count levels, with entry $(\sqrt{r_m} - \sqrt{r_n})^2 G_{mn}$. If B retains all six literal units, the actual unit sum further gives $G_{mn} = 0$ unless $Nm \equiv Nn \pmod{36}$. These are exact deletions, not a power saving. Even zero different-level pairings leave the nonnegative within-level energies $\sum_k (k-1)^2 Q_B(\mu \mathbf{1}_{r=k}, \mu \mathbf{1}_{r=k})$. The unequal-product coupled fourth moment and its mixed Möbius term remain unestimated. No new analytic exponent, energy lower bound, Lean formalization or GRH conclusion follows.

19 Proof and arithmetic boundary of the covariance budget

19.1 The positive-contraction calculation

Since G is the exact raw Gram, it is positive semidefinite. If $\Lambda_B > 0$, put $T = 2G/\Lambda_B - I$, a self-adjoint contraction. Then

$$\Re(v^*RGv) = \frac{\Lambda_B}{2}\{M_1 + \Re(v^*RTv)\} \geq \frac{\Lambda_B}{2}\{M_1 - \|Rv\|_2\|v\|_2\},$$

which is (28); $\Lambda_B = 0$ is immediate. The Hermitian rank-two matrix $\{(Rv)v^* + v(Rv)^*\}/2$ has possible nonzero eigenvalues $(M_1 \pm \sqrt{M_0M_2})/2$. Its negative spectral projection realizes the lower bound among all $0 \leq G \leq \Lambda_B I$, a linear-algebra sharpness statement only. If $a \leq r_m \leq b$ on the weighted support, $a + b > 0$, the pointwise inequality $r_m^2 \leq (a+b)r_m - ab$ also gives

$$\Re Q_B(C_U, \mu) \geq -\frac{\Lambda_B M_0 (b-a)^2}{8D(a+b)}.$$

For $a = b = 0$ the covariance is zero. The constant follows by maximizing $\sqrt{(a+b)t - ab} - t$ on $[a, b]$.

19.2 The balanced-divisor second moment

Set $T_0 = \sqrt{D}$. The high-window factor restrictions imply

$$\sqrt{2/7}\,T_0 < Na, Nb < \sqrt{14}\,T_0.$$

The elementary ideal count $\#\{a : Na \leq Y\} \ll_K Y$ proves the first assertion of (29). For the second, two ordered factorizations $ab = cd = m$ have the unique form

$$a = gh, \quad c = gk, \quad b = k\ell, \quad d = h\ell, \qquad g = (a, c).$$

Squarefreeness makes g, h, k, ℓ pairwise coprime and squarefree. To upper-bound this positive count, enlarge to arbitrary ideals. For $M = \max(Nh, Nk)$ the individual bounds are

$$Ng \leq \frac{\sqrt{14}\,T_0}{M}, \qquad N\ell \leq \frac{\sqrt{14}\,T_0}{M}, \qquad 1 \leq M \leq \sqrt{14}\,T_0.$$

On $2^j \leq M < 2^{j+1}$ there are $O_K(2^{2j})$ choices of (h, k) and $O_K((T_0/2^j)^2)$ choices of (g, ℓ), with bounded terminal ranges absorbed in the constant. Each dyadic range contributes $O_K(D)$; there are $O(\log(2D))$ ranges. This proves the second moment with no extra logarithm from counting only $NgN\ell$. Restricting to primes outside fixed S decreases the count. Hence

$$M_0, M_1 \ll_K \|W\|_\infty^2 D, \qquad M_2 \ll_K \|W\|_\infty^2 D \log(2D).$$

These constants are independent of the cutoff in the stated window. Only the common supremum norm is used, including for complex W.

19.3 What the commutator actually controls

Write

$$t_{mn} = D^{-1}\Re\{\lambda_m\overline{\lambda_n}G_{mn}\}, \qquad \delta_{mn} = (\sqrt{r_m} - \sqrt{r_n})^2.$$

The exact commutator identity proves

$$\Re Q_B(C_U, \mu) \geq Q_B(\sqrt{r}\,\mu, \sqrt{r}\,\mu) - \tfrac{1}{2}\sum_{m,n}\delta_{mn}|t_{mn}|.$$

The displayed loss retains the actual character and profile phases before taking absolute values; it is at most the phase-blind loss obtained from $|t_{mn}| \leq D^{-1}|\lambda_m\lambda_n G_{mn}|$. On the accepted four-column smooth B_7 fixture at amplitude $x \geq 0$, the positive term is $18/25$, this coefficient-adapted loss is $9x/50$, and the lower bound is attained. Purely imaginary entries of the full complex Gram make the phase-blind loss larger. This calibrates the new budget and does not give an asymptotic obstruction.

For the unit deletion let $\zeta_6 = 1+\omega$. Every good prime norm q is $1 \pmod 6$, and direct residue-field exponentiation gives $x_p(\zeta_6) = \zeta_6^{(q-1)/6}$. Multiplying over $p \mid m$ and reducing $Nm = \prod_{p|m}(1+6e_p)$ modulo 36 gives

$$x_m(\zeta_6) = \zeta_6^{(Nm-1)/6 \bmod 6}.$$

Multiplication of all literal rows by ζ_6 preserves their raw zeros and proves the norm-congruence deletion in the core. The vector budget may be applied separately on these six classes. No D-dependent number of classes or conductor saving is produced.

The moment proof above counts the coefficient diagonal $ab = cd$. It has not estimated the unequal-product character quadruples in the coupled row fourth moment. Replacing the inherited DX norm by $D^{1-\sigma}X$ merely because equal-level and different-class entries vanish is the unsupported step. Small negative commutator loss alone also leaves the within-level energies in the core. Principal residues are unchanged; this covariance is used on the nonprincipal blocks where the original residue is zero, not as a new centering prescription.

19.4 The complete near-saturating face

For a six-factor row block let $R_0 = \prod_{j=1}^{6} A_j$, $M = \max(A_1, A_2)$ and $\prod_j A_j^j \leq H$. The accepted exact-mask block norm is

$$\Lambda_A \ll_{\nu,S,\epsilon} (HD)^{\epsilon}\{R_0 + DR_0/M + D^{2/3}R_0/M^{1/3}\}.$$

The column moment and covariance proofs apply to every such block. Write $x_j = \log A_j$, $h = \log H$, $c_0 = \sum_j jx_j$ and $r_0 = \sum_j x_j$. Then, for $x_1 \geq x_2$,

$$h - 3(r_0 - \max(x_1, x_2)) = h - c_0 + (x_1 - x_2) + x_4 + 2x_5 + 3x_6;$$

for $x_2 \geq x_1$ replace $x_1 - x_2$ by $2(x_2 - x_1)$. All terms are nonnegative. Thus the middle monomial is saturated on

$$A_1 = A_2, \qquad A_1A_3 = H^{1/3}, \qquad A_4 = A_5 = A_6 = 1,$$

with full capacity. The balanced $A_3 = 1$ case is only an endpoint. The a_3 direction remains nonprincipal by its outside-S valuation $3 \pmod 6$. If $R_0/M \geq H^{1/3}D^{-\eta}$, the total displayed slack is at most $3\eta \log D$. Therefore a fixed power gain must be uniform on this full neighboring region. The proved budget still costs DR_0/M; it gives no global improvement over the accepted envelope.

The independent standard-library diagnostic checks nine ideal moment cases, eight raw Gram cases with 27 000 phase values and 5 184 raw zeros, exact radical commutator/vector budgets,

and 227 exponent cases. Finite tables use common norm profiles realizable by smooth disjoint bumps; they are not tests of an infinite analytic estimate. Full derivation, source bindings and preserved checks are under `planning/covariance-budget-20261105/attempt/` and `outputs/checks/2026-11-05-covariance-budget/attempt/`.

Classical reference NIST Digital Library of Mathematical Functions, Section25.15, Dirichlet L-functions, equations 25.15.5–6; primitive functional equation and finite Gauss sums (snapshot pinned 4 November 2026). The normalization and explicit finite Gauss sums used here are stated above.